

...: ATOMIC STRUCTURE :...

7/08/24

LN-01

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DEFINITION : The branch of chemistry which deals with the structure of atom

SUB-TOPICS :

→ Dalton's atomic theory

→ Discovery of fundamental particles (e^- , proton, neutron)

→ Arrangement of fundamental particles, (Atomic model)

→ Thomson atomic Model

→ Rutherford atomic Model.

→ Bohr's Model

→ Wave mechanical model of atom

→ Dual nature of light.

→ Planck's Quantum theory

→ De-Broglie hypothesis

→ Heisenberg's uncertainty principle

* → Hydrogen spectrum

* → Quantum numbers.

DALTON'S ATOMIC THEORY :→ (1808) (DAT)

POSTULATES OF DALTON'S ATOMIC THEORY:→

① Each element is composed of small particles called atoms.

Atom is a Greek word and it means indivisible i.e. an ultimate particle that cannot be further subdivided.

② Atoms of a particular element are like but differ from atoms of other elements.

③ Atom of each element is an ultimate particle and it has a characteristic mass but it is structureless. Atoms can neither be created nor destroyed.

④ Atoms of different elements take part in a chemical rxn to form molecules.

⑤ Atoms always combine in a simple ratio.

34 DEMERITS OF DALTON'S ATOMIC THEORY: →

① It does not account for subatomic particles.

Dalton's atomic theory stated that atoms are indivisible. However, the discovery of subatomic particles (e⁻, p, n) disproved this postulate.

② It does not account for isotopes.

This theory states that atoms of the same type having similar properties in all respects. But the isotopes of an element having different masses will not follow this theory.

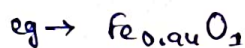
③ It does not account for isobars.

This theory states that masses of atoms of different elements must be different. However, it is possible for two different elements having the same mass number.

④ It does not account for allotropes.

The difference in the properties of diamond and graphite shows carbon having different allotropic forms, which disproved DAT.

⑤ Elements need not to combine in a simple ratio always.



MERITS OF DAT: →

① The laws of multiple proportion, The laws of conservation of mass,

The laws of constant proportion are not violated by DAT.

② This theory provides a basis of difference between elements & compounds.

Q Which of the following laws of chemical combination is not proved by Dalton.

(A) Law of Multiple proportion

(B) Law of constant proportion

(C) Law of conservation of mass

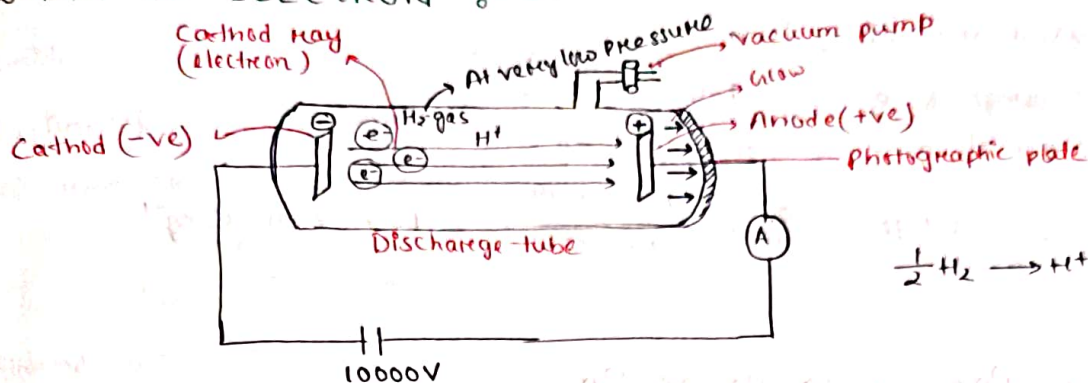
(D) Avogadro's law.

DISCOVERY OF FUNDAMENTAL PARTICLES :->

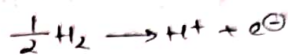
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1. DISCOVERY OF ELECTRON :-



Cathod ray
Cathod ray is made
of small particles
H2 gas breaks into
into e- and H+
these particles are

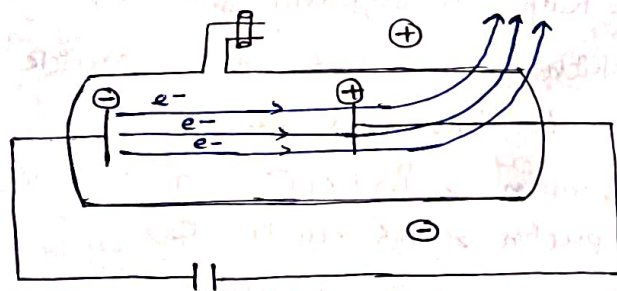


-> Julius Plucker was studying the conductivity of gases and he found that normally gases are bad conductors of electricity but when he created very low pressure (10^{-4} ATM) in the discharged tube and applied very high voltage (10^4 Volt or more) than he found that gases start conducting electricity because they ionised. In this condition some sort of invisible rays moved from cathode to anode. Since these rays appear to be originating from cathode. Hence they are named as cathode rays.

In 1897 J.J. Thomson named the particle as electron.

● Characteristics of Cathode rays :->

- ① Cathode rays travel in straight line with very high velocity and produces shadow of the object placed in their path
- ② Cathode ray produce mechanical effect i.e. if a small paddle wheel is placed in their path it starts rotating which indicates that they consist of material part (mass)
- ③ When electrical and magnetic field are applied to cathode ray in discharge tube then rays are deflected towards +ve field of the plate which shows that cathode ray consist of -vely charged particles called electrons.



- ④ Cathode rays create fluorescence.
- ⑤ Cathode rays penetrates through thin sheets of Aluminium and metal.

- ⑥ The ratio of charge to mass is known as specific charge. 86
 Specific charge $\boxed{e/m}$ is same for all cathod rays irrespective of the gas used in that tube.

$$\text{charge of } e^- = 1.6 \times 10^{-19} \text{ C}$$

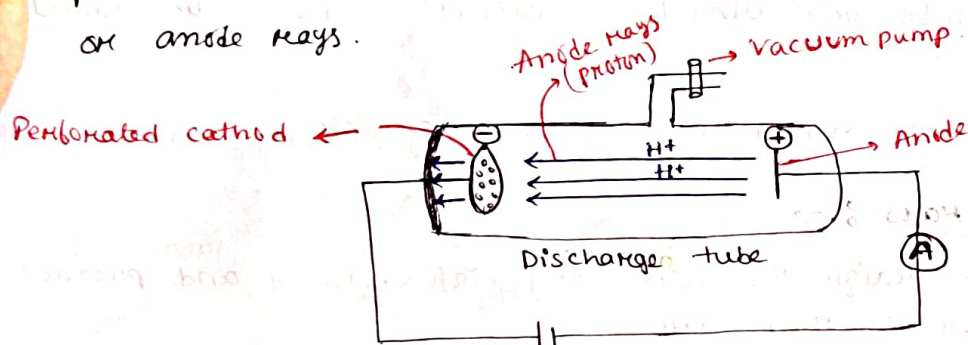
$$\text{Mass of } e^- = 9.1 \times 10^{-31} \text{ kg}$$

$$\text{Specific charge} = \frac{e}{m} = \frac{1.6 \times 10^{-19}}{9.1 \times 10^{-31}} = 1.76 \times 10^{11} \text{ C kg}^{-1}$$

2. DISCOVERY OF PROTON OR ANODE RAYS: →

→ Goldstein. (1886)

→ In 1886, Goldstein performed the experiment using perforated cathode. (having holes or canals) and found that rays are not only travelling from cathode to anode but simultaneously also travelling from anode to cathode and passes through holes or canals of cathode and these are termed as canal rays or anode rays.



⊙ Characteristics of Anode Rays : →

① They deflect towards -ve plate of the field which indicates that they are +vely charged.

② $\frac{e}{m}$ ratio for anode rays is smaller than that of cathode rays.

③ $\frac{e}{m}$ depends upon nature of gas taken in the tube which indicates that +ve particle consisting +ve rays is different for different gas. This $\frac{e}{m}$ is maximum for hydrogen which shows that +ve particle obtain from H_2 gas is lightest and this particle is known as proton.

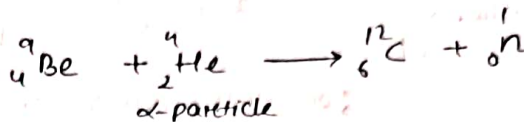
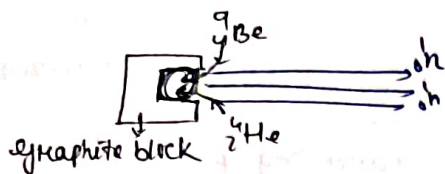
$$\text{Mass of proton} = 1.67 \times 10^{-24} \text{ kg}$$

$$\text{charge of proton} = 1.6 \times 10^{-19} \text{ C}$$

$$\frac{e}{m} \text{ for proton} = \frac{e}{m} = \frac{1.6 \times 10^{-19}}{1.67 \times 10^{-24}} = 9.5 \times 10^7 \text{ C kg}^{-1}$$

3. DISCOVERY OF NEUTRON :->

-> In 1932, Chadwick was bombarding sheet of Beryllium (Be) with α -particles and he found that a new particle which is electrically neutral but having slightly greater mass than proton are emitted and he named this particle as neutron.



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• Electrons, protons, neutrons are called fundamental particles

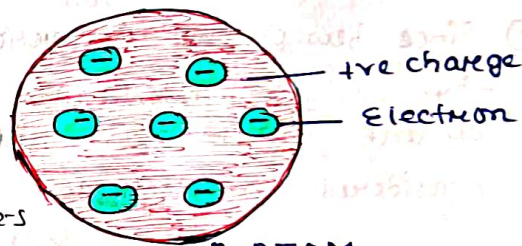
Particle	Symbol	Mass	Charge
Electron	${}_1^0\text{e}$	$9.1 \times 10^{-31} \text{ kg}$	$1.602 \times 10^{-19} \text{ C}$
Proton	${}_1^1\text{H}$	$1.673 \times 10^{-27} \text{ kg}$	$1.602 \times 10^{-19} \text{ C}$
Neutron	${}_0^1\text{n}$	$1.675 \times 10^{-27} \text{ kg}$	0

ATOMIC MODELS :->

THOMSON'S ATOMIC MODEL :-> (1904)

-> Thomson was the first to propose a detailed model of atom.

Thomson proposed that an atom consists of a uniform sphere of +ve charge in which there are distributed more or less uniformly.

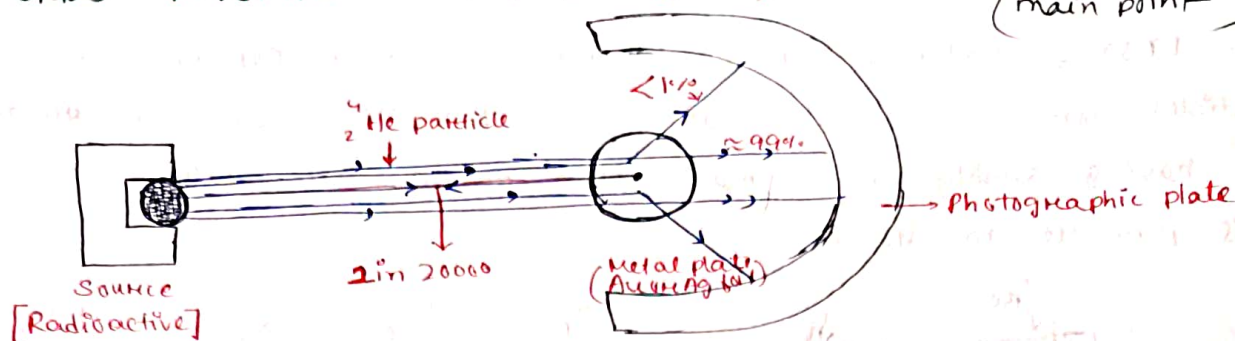


-> The model of atom is known as plum pudding model or watermelon model or choco cookies model.

Drawbacks :-

- ① An imp. drawback of this model is that the mass of atom is considered to be evenly spread over the atom.
- ② It is a static model. It does not reflect the movement of electron.
- ③ It could not explain the stability of an atom.

RUTHERFORD'S ATOMIC MODEL $\rightarrow$ (1913) $\rightarrow$ concept of nucleus (main point)



RUTHERFORD'S SCATTERING EXPERIMENT

$\rightarrow$ Rutherford's experiment is also called " α -particle scattering experiment"

Observations :-

- ① Most of the α -particle passed through the gold foil undeflected.
- ② A small fraction of α -particle were deflected by small angles.
- ③ A very few α particles 1 in 20,000 bounced back, i.e., were deflected by nearly 180° .

Conclusions :-

- ① Since most of the α particles went straight through the metal foil undeflected, it means that there must be a very large empty space within the atom.
- ② Since few of the α particles were deflected from their original paths through moderate angles, it was concluded that there must be some particles present which was hit by α -particles. Later on, these particles are considered as electrons.
- ③ A very few α -particles suffered from deflections or even returned on their path indicating that the nucleus is rigid and α -particles recoil due to direct collision with the heavy, +vely charged mass.

Some imp. points :-

- ① An atom consist of a heavy +vely charged nucleus where all the protons and neutrons are present. Protons and neutrons are collectively called as nucleons.
 - ② Almost whole of the mass of the atom is contributed by these nucleons.
 - ③ The magnitude of the +ve charge of the nucleus is different in different atoms.
 - ④ Volume of the nucleus is very small and is only a minute fraction of the total volume of the atom.
- Nucleus has the diameter of the order of 10^{-12} to 10^{-13} cm and an atom has diameter of the order of 10^{-8} cm.

$$\frac{D_A}{D_N} = \frac{\text{Diameter of Atom}}{\text{Diameter of Nucleus}} = \frac{10^8}{10^{-13}} = 10^5$$

$$D_A = 10^5 D_N$$

nucleus contribution $\rightarrow$
99.99% mass of the atom
is the mass of nucleus.

- ③ There is an empty space around the nucleus called extra nuclear part. In this part e^- are present. The no. of e^- in an atom is always equal to no. of protons present in the nucleus. As the nuclear part of atom is responsible for the mass of the atom, the extra nuclear part is responsible for the volume of the atom.

Volume Ratio of atom and Nucleus

$$\text{Radius of atom} = 10^{-10} \text{ cm}$$

$$\text{Radius of nucleus} = 10^{-15} \text{ cm}$$

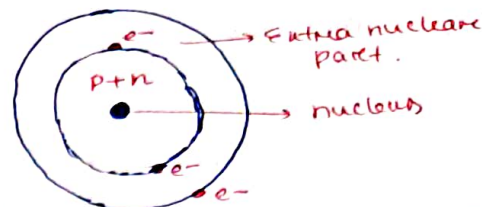
$$\text{Volume of atom} \Rightarrow \frac{4}{3} \pi R^3 \Rightarrow \frac{4}{3} \pi \cdot 10^{-30} \text{ cm}^3$$

$$\text{Volume of nucleus} \Rightarrow \frac{4}{3} \pi r^3 = \frac{4}{3} \pi \cdot 10^{-45} \text{ cm}^3$$

$$\frac{V_A}{V_N} = \frac{\frac{4}{3} \pi 10^{-30}}{\frac{4}{3} \pi 10^{-45}} = 10^{15}$$

$$V_A = 10^{15} V_N$$

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$\rightarrow$ The radius of the nucleus $\propto \sqrt[3]{A}$ of the nucleus (mass no. of ^AH)

Radius of any nucleus $\propto$

$$R \propto A^{1/3}$$

$$R = R_0 A^{1/3}$$

A = mass no.

R = Radius of nucleus

$$R_0 \approx \text{const.} \approx 1.1 \times 10^{-13} \rightarrow 1.4 \times 10^{-13} \text{ cm}$$

$$\approx \frac{4}{3} \times 10^{-13} \text{ cm}$$

Q Calculate the radius of Al -atom.

Soln:

$$R = \frac{4}{3} \times 10^{-13} \times (27)^{1/3} = \frac{4}{3} \times 10^{-13} \times 3^{1/3} = 4 \times 10^{-13}$$

- ④ no. of α -particles scattered at any angle θ is inversely proportional to $\sin \frac{\theta}{2}$.

$$N(\theta) \propto \frac{1}{(\sin \frac{\theta}{2})^4}$$

$N(\theta)$ = no. of α -particles scattered at an angle θ

Higher value of $\theta \rightarrow$ less α -particles

Lower value of $\theta \rightarrow$ More α -particles

Q 96 no of α -particles scattered at an angle 60° at 112 per minute. 90
 - then calculate no. α particles scattering at an angle of 90° per min Ans = 2.8

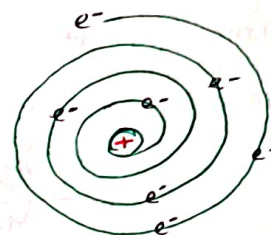
Soln:-

$$\begin{aligned} \frac{N_1}{N_2} &= \frac{(\sin \frac{\theta_1}{2})^4}{(\sin \frac{\theta_2}{2})^4} \\ \Rightarrow \frac{112}{N_2} &= \frac{(\sin 45^\circ)^4}{(\sin 30^\circ)^4} \\ \Rightarrow \frac{112}{N_2} &= \frac{(\frac{1}{\sqrt{2}})^4}{(\frac{1}{2})^4} \\ \Rightarrow \frac{112}{N_2} &= \frac{1/4}{1/16} \\ \Rightarrow N_2 &= \frac{112}{4} = 2.8 \end{aligned}$$

DRAWBACKS :-

→ This theory could not explain the stability of atom.

A/c to Maxwell e^- loose its energy continuously in the form of electromagnetic radiations. As a result of this the e^- s should loose energy at every turn and move closer and closer to the nucleus following a spiral path. The ultimate result will be that the e^- will fall into the nucleus thereby making the atom unstable.

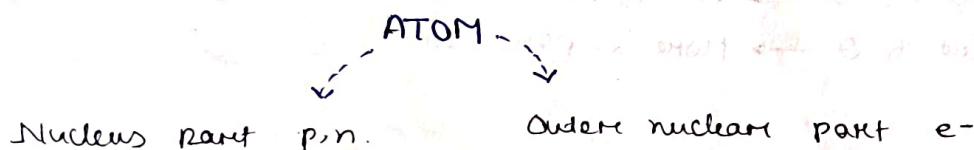


As the e^- loose energy continuously the observed spectrum should be continuous but the actual observed spectrum consist of well defined lines of definite frequencies. Hence the loss of energy of e^- s is not continuous in an atom.

ATOM :-

ATOMIC NUMBER :- It is represented by 'Z'. The no of protons present in the nucleus is called Atomic no. of an element.

MASS NUMBER :- It is represented by 'A'. The sum of no. of neutrons and protons is called mass no. of the element. It is also known as no of nucleons because neutrons and protons are present in nucleus.



Atom representation



$Z \Rightarrow$ Atomic number $\Rightarrow$ no. of proton

for neutral atom,

no. of proton = no. of electron.

$A \Rightarrow$ Mass number $\Rightarrow$ no. of p + no. of n

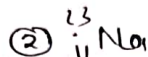
Q Calculate the e⁻, p, n no. of the following.



no. of e⁻ = 2

no. of p = 2

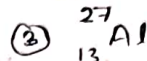
no. of n = 4 - 2 = 2



no. of e⁻ = 11

no. of p = 11

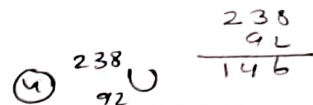
no. of n = 23 - 11 = 12



no. of e⁻ = 13

no. of p = 13

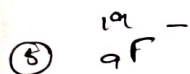
no. of n = 27 - 13 = 14



no. of e⁻ = 92

no. of p = 92

no. of n = 238 - 92 = 146



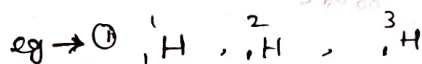
no. of e⁻ = 9 - (-1) = 10

no. of p = 9

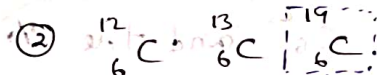
no. of n = 10

SOME IMP DEFINITIONS $\Rightarrow$

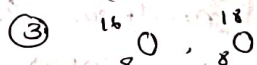
1. **ISOTOPS** $\Rightarrow$ They are atoms of a given element which have the same atomic no. but differ in their mass no.



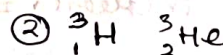
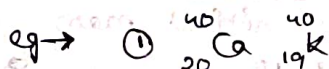
Protium Deuterium Tritium $\rightarrow$ lightest radioactive isotope.



$\downarrow$ radioactive isotope



2. **ISOBAR** $\Rightarrow$ Isobars are the atoms of different element which have the same mass no. but different atomic no. i.e. they have different no. e⁻, p, n but same no. of neutrons and protons remains same.



3. **ISODIAPHER** $\Rightarrow$ They are the atoms of different element which have the same difference of no. of neutrons and protons.

$[(n-p) \text{ same}]$



p = 5

n = 6

n - p = 1



p = 6

n = 7

n - p = 1

} isodiaphere.



p = 7

n = 8

n - p = 1



p = 9

n = 10

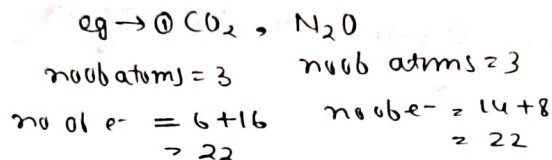
n - p = 1

4. **ISOTONES** $\rightarrow$ They are the atoms of different element which have the same no. of neutrons. 9L

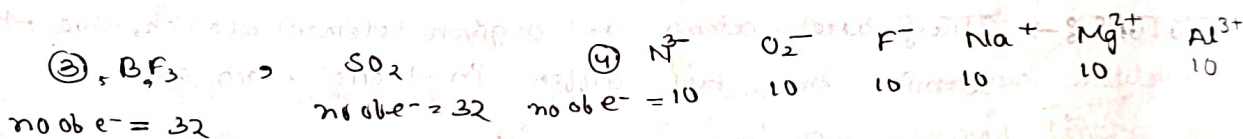
(**Isoneutronic**
OR
Isotonic)



5. **ISOSTERS** $\rightarrow$ They are the molecules which have the same no. of atoms and electrons.



6. **ISOELECTRONIC SPECIES** $\rightarrow$ They are the atoms, molecules and ions which have the same no. of e^- s



Q If the mass of neutron is doubled and mass of e^- is halved then find out the new atomic mass of ${}^{12}_6\text{C}$ and the % by which it is increased.

Soln:- Initially,

$e^- \rightarrow 6 \text{ amu}$
 $p \rightarrow 6 \text{ amu}$
 $n \rightarrow 6 \text{ amu}$ } 12 amu

finally, $e^- \rightarrow \frac{6}{2} = 3 \text{ amu}$

$p \rightarrow 6 \text{ amu}$
 $n \rightarrow 6 \times 2 = 12 \text{ amu}$ } 18 amu.

$$\% \text{ increased} = \frac{\text{Final mass} - \text{Initial mass}}{\text{Initial mass}} \times 100$$

$$= \frac{18 - 12}{12} \times 100 = \frac{6}{12} \times 100 = 50\%$$

Q If mass of n is doubled, mass of proton is halved and mass of e^- is doubled, then new atomic mass will be — also find the % increase. $({}^nC_6)$

Solns - Initially.

$$\begin{aligned} \text{no of } p &= 6 \text{ amu} \\ n &= 6 \text{ amu} \\ e &= 6 \text{ amu} \end{aligned} \quad \left. \vphantom{\begin{aligned} \text{no of } p &= 6 \text{ amu} \\ n &= 6 \text{ amu} \\ e &= 6 \text{ amu} \end{aligned}} \right\} 12 \text{ amu}$$

finally

$$\begin{aligned} p &= 6_2 = 3 \text{ amu} \\ n &= 2 \times 6 = 12 \text{ amu} \\ e &= 6 \text{ amu} \end{aligned} \quad \left. \vphantom{\begin{aligned} p &= 6_2 = 3 \text{ amu} \\ n &= 2 \times 6 = 12 \text{ amu} \\ e &= 6 \text{ amu} \end{aligned}} \right\} 15 \text{ amu}$$

$$\therefore \text{ Increased } = \frac{15-12}{12} \times 100$$

$$= \frac{3}{12} \times 100 = 25\%$$

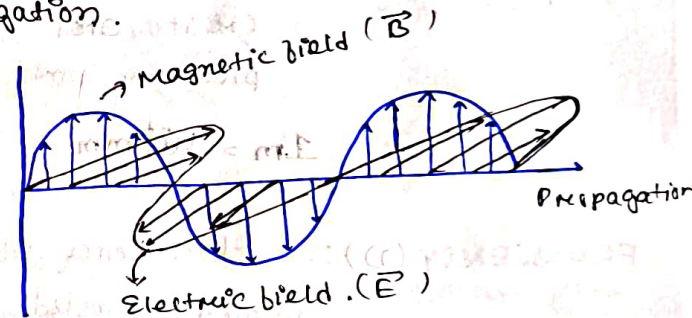
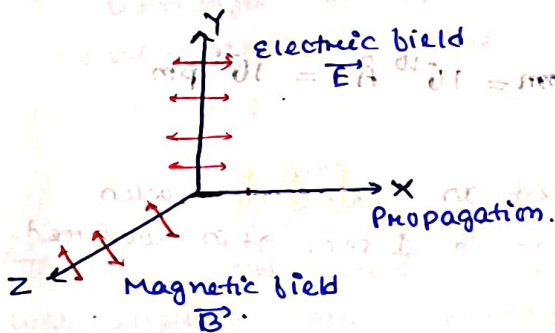
Q no of p in X^{2-} in 16 then no. of e^- in X^{12} will be.

$$\begin{aligned} \text{Soln: } X^{2-} &\rightarrow \text{no of } p = 16 \\ \text{no of } e^- &= 16 - 2 = 14 \end{aligned}$$

19-08-24
L-06

ELECTRO MAGNETIC WAVES . OR RADIATION :-

→ wave in which both electrical and magnetic field are present are known as electromagnetic wave. Electric field and magnetic field are $\perp$ to each other as well as they are also $\perp$ to the direction of propagation of wave. EMW always travel with speed of light and they don't require any medium for propagation.



→ A/c to this theory the energy is transmitted from one body to another in the form of waves and these waves travel in the space with the speed of light. These waves are known as EMWs or Radiant energies.
eg → Radio wave, Microwave etc.

PROPERTIES →

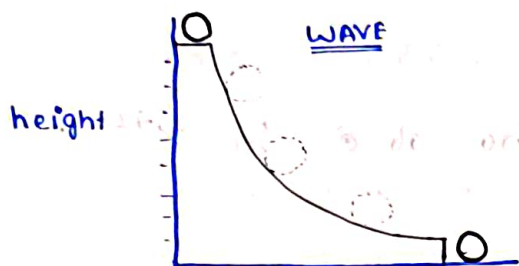
- ① The radiant energy don't need any medium for propagation.
- ② The radiant energy have electric and magnetic field and travel at right angle to these fields.

DUAL NATURE OF LIGHT : →

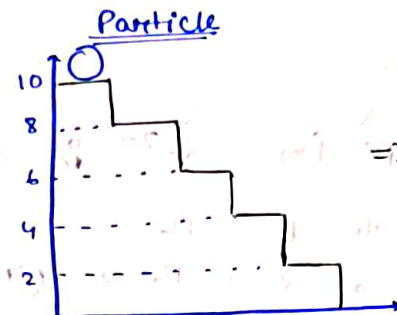
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→ Light possesses dual nature i.e., it has both particle as well as wave nature.

The basic difference b/w wave and particle in - waves are continuous but particles are discontinuous in nature.



Wave value $\rightarrow \infty$



$\Rightarrow$ discontinuous
value should be finite.

→ It is a mode of transferring energy without transferring any matter.

GENERAL TERMS RELATED TO WAVE : →

WAVELENGTH (λ) : → It is defined as the distance b/w two nearest crest or trough. It is measured in $\text{\AA}$ (Angstrom), picometer (pm), nm, cm etc.

$$1\text{m} = 10^3\text{mm} = 10^6\mu\text{m} = 10^9\text{nm} = 10^{10}\text{\AA} = 10^{12}\text{pm}$$

FREQUENCY (ν) : → Frequency of wave is defined as the no. of waves which passed through a point in 1 sec. It is measured in terms of Hertz or s^{-1} .

WAVE NUMBER ($\bar{\nu}$) : → It is the reciprocal of wavelength i.e. no. of waves present in unit length.

$$\bar{\nu} = \frac{1}{\lambda}$$

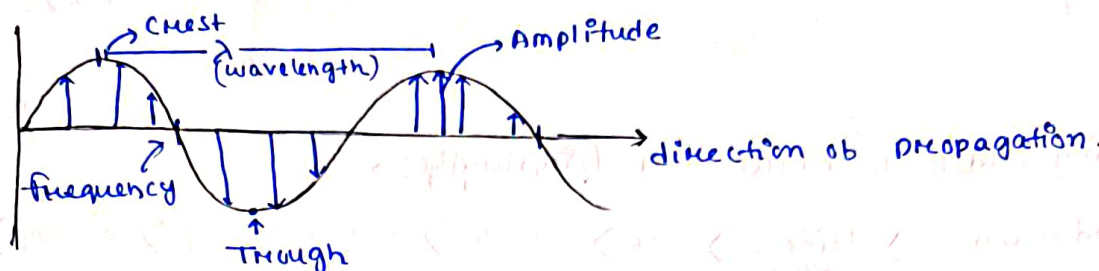
TIME PERIOD (T) : → The time taken by a wave to pass through 1 point.

$$T = \frac{1}{\nu}$$

$$T = \frac{\text{distance travelled}}{\text{Velocity}}$$

VELOCITY (v) : → Velocity of a wave is defined as distance covered by a wave in one second.

AMPLITUDE (α): → The amplitude of a wave is defined as the height of crest or trough.



RELATION BETWEEN VELOCITY, FREQUENCY, WAVELENGTH: →

$$c = v\lambda \quad \lambda \propto \frac{1}{v}$$

Q. If wavelength of a particular wave is $5000 \text{ \AA}$ then calculate its frequency.

Soln:-

$$c = v\lambda$$

$$v = \frac{c}{\lambda} = \frac{3 \times 10^8}{5000 \times 10^{-10-7}} = \frac{3}{5} \times 10^{15} = 0.6 \times 10^{15} = 6 \times 10^{14}$$

Q. If wavelength of visible light varies from $4000 - 8000 \text{ \AA}$ then calculate frequency range of visible light.

Soln:-

$$\lambda_1 = 4000 \quad \lambda_2 = 8000 \text{ \AA}$$

$$v_1 = \frac{c}{\lambda_1} = \frac{3 \times 10^8}{4000 \times 10^{-10}} = 10^{15} \times 0.75 = 7.5 \times 10^{14}$$

$$v_2 = \frac{3 \times 10^8}{8000 \times 10^{-10}} = 0.375 \times 10^{15} = 3.75 \times 10^{14}$$

Q. The hiral radio station broadcast a frequency of 1368 KHz . Calculate the wavelength of EMW emitted by the transmitter.

Soln:-

$$\lambda = \frac{c}{v}$$

$$= \frac{3 \times 10^8}{1368 \times 10^3} = \frac{1}{456} \times 10^8 = 219.3 \text{ metre.}$$

Q. Calculate $\bar{\nu}$ in cm^{-1} and v of yellow radiation having a wavelength of $5800 \text{ \AA}$.

$$\bar{\nu} = \frac{1}{\lambda}$$

$$v = \frac{3 \times 10^8}{5800 \times 10^{-10}} = 0.051 \times 10^{16} = 5.1 \times 10^{14}$$

$$\bar{\nu} = \frac{1}{5800 \times 10^{-8}} = 17214.37$$

Q - How long it would take a radio wave of frequency $6 \times 10^3 \text{ sec}^{-1}$ to travel from mars to earth, that is a distance of $8 \times 10^7 \text{ km}$ 4 min 26 sec

Soln:- $t = \frac{S}{V} = \frac{8 \times 10^7 \times 10^3}{3 \times 10^8} = \frac{800}{3} = 266.67 = 4 \text{ min } 26 \text{ sec}$

Decreasing order of EMW with Wavelength $\Rightarrow$

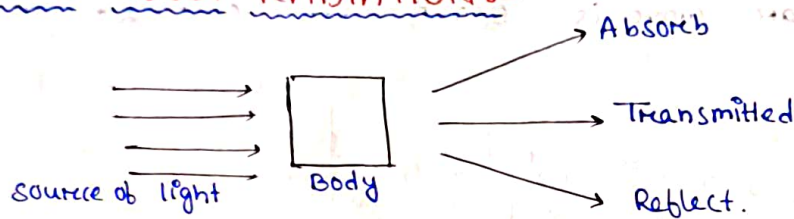
Radio waves > Micro waves > IR > visible > UV > X-ray > γ -ray > Cosmic ray.

frequency $\uparrow$
 $\lambda \downarrow$

VI B G Y O R X
PO Y G B I V V
 $\lambda \downarrow$

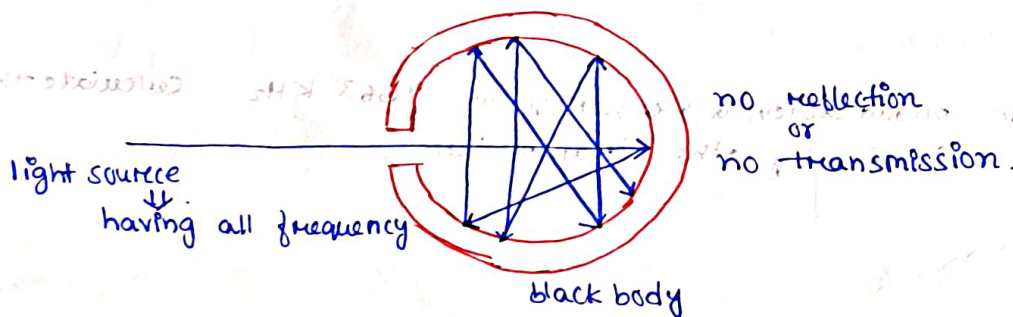
heat
Red colour
Yellow/white
Blue
Dibb 2
discontinuous
(particle nature)

BLACK BODY RADIATION



The colour change of iron rod cannot be explained by wave nature.

$\rightarrow$ A black body is an ideal surface or body, that completely absorbs or re-emits all radiations that falls on it without deflection. Absorption of radiation is independent of the direction of incidence, frequency and wavelength of radiation.



PROPERTIES OF BLACK BODY $\Rightarrow$

- ① It absorbs all the incident radiation irrespective of wavelength and frequency range.
- ② There would be no radiation reflected or transmitted at normal temp.
- ③ An object which is a good absorber always be a good emitter. Therefore a perfect black body will also be a good emitter.

BLACK BODY RADIATION :->

→ EM radiation emitted by any heated object is called black body radiation. It is also called cavity radiation.

Depending on their temp all object generates ^{EM} radiation.

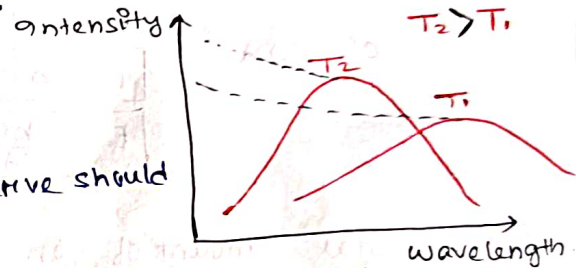
Colder objects emit low frequency waves such as radio and micro.

Heated objects emits high frequency waves.

BLACK BODY SPECTRUM :->

One of the characteristic feature of black body radiation is a continuous frequency spectrum called as black body spectrum. It is completely independent on temp.

• A graph is obtained by plotting the intensity of radiation against wavelength gives the following details.



- ① The nature of radiation depends upon the temp of black body.
- ② As the energy emitted is continuous the curve should be as shown by the dotted lines.
- ③ At a given temp the intensity of radiation increases with the wavelength reaches max and then decreases.
- ④ The intensity of radiation is greatest at the medium wavelengths and least at highest and lowest wavelength.
- ⑤ As the temp increases the peak of max intensity shifts towards the shorter wavelengths.

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PARTICLE NATURE OF LIGHT :->

→ There are certain phenomena like interference, diffraction etc which can be explained on the basis of wave nature of light. There are certain phenomena like black body radiation, photoelectric effect which can only be explained on the basis of particle nature of light.

PLANK'S QUANTUM THEORY :->

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→ Plank studied the radiation emitted from black body and to explain the black body radiation he gave quantum theory.

Assumptions:-

- ① The radiant energy emitted or absorbed by a body is not continuous but discontinuous in the form of small discrete packets of energy and these packets are called quantum.
- ② In case of light the smallest packet of energy is called a PHOTON but in general the smallest packet of energy is called a QUANTUM.
- ③ The energy of each quantum is directly proportional to frequency of radiation i.e.

$$c = \nu \lambda$$

$$E \propto \nu$$

$$E = h\nu$$

$$E = \frac{hc}{\lambda}$$

h = Planck's constant

$$= 6.626 \times 10^{-34} \text{ JS}$$

$$E = \frac{hc}{\lambda} = \frac{12400}{\lambda(\text{\AA})}$$

$$= \frac{1240}{\lambda(\text{nm})}$$

- ④ The total amount of energy transmitted from one body to another will be sum integral multiple of energy of a quantum

$$E = nh\nu = \frac{nhc}{\lambda} = nhc\bar{\nu}$$

• Energy of one photon i.e. for quantum $E = h\nu$

• Energy of more than one photon or n no. of photon or n Quanta

$$E = nh\nu$$

Q Calculate energy of photon having wavelength 3000 Å

Soln:- $E = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \text{ Js} \times 3 \times 10^8 \text{ m}}{3000 \times 10^{-10} \text{ m}}$ OR. $\frac{12400}{3000} = 4.1 \text{ eV}$

$$= \frac{6.63 \times 10^{-19} \text{ J}}{1.6 \times 10^{-19}}$$

$$= 4.1 \text{ eV}$$

$$1 \text{ eV} = 96 \text{ kJ mol}^{-1}$$

Q Calculate energy of a photon having frequency 2×10^{15} Hz.

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Soln:- $E = h\nu = 6.626 \times 10^{-34} \times 2 \times 10^{15} = \frac{13.2 \times 10^{-19}}{1.6 \times 10^{-19}} = 8.2 \text{ eV}$

Q Calculate energy of one mole of photon having wavelength $4100 \text{ \AA}$.

Soln:- $E = \frac{nhc}{\lambda} = \frac{12400}{4100} = 3.02 \text{ eV}$

Q Calculate no of photons having wavelength 310 nm which are required to obtain energy of $2 \times 10^5 \text{ J}$

Soln:- $2 \times 10^5 = n \frac{nhc}{\lambda}$ $\therefore n = \frac{E\lambda}{hc} = \frac{2 \times 10^5 \times 310 \times 10^{-9}}{6.62 \times 10^{-34} \times 3 \times 10^8}$
 $= 3.175 \times 10^{23}$
 $n = \frac{10^5}{2} = \frac{5 \times 10^4 \times 96}{1.6 \times 10^{-19}} = 3 \times 10^{23}$

Q Calculate the energy of a photon of sodium light of wavelength $5.862 \times 10^{-16} \text{ m}$ in J. ($3.38 \times 10^{-10} \text{ J}$)

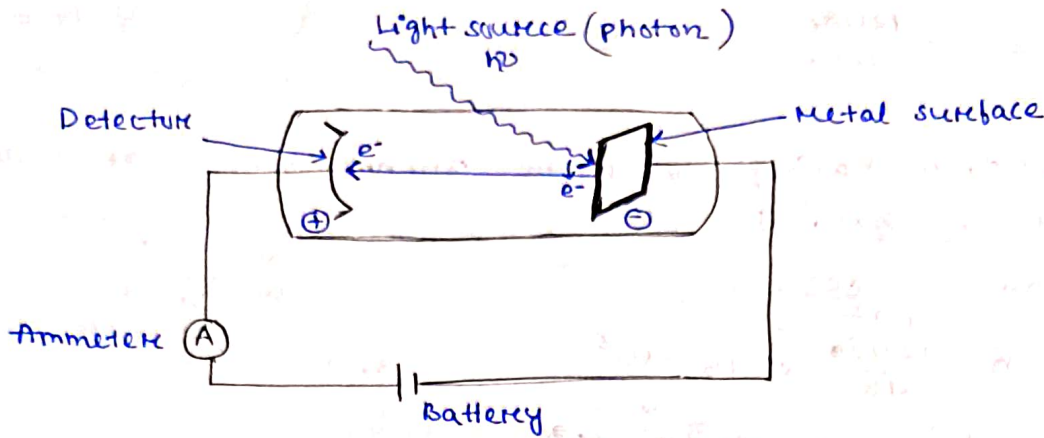
Soln:- $E = \frac{hc}{\lambda} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{5.862 \times 10^{-16}} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{5.862 \times 10^{-16}} = 3.38 \times 10^{-10} \text{ J}$

Q Calculate the λ and ν of a photon having an energy of 2 eV .

Soln:- $\lambda = \frac{12400}{2} = 6200 \text{ \AA}$
 $= 6200 \times 10^{-10} \text{ m}$
 $= 6.2 \times 10^{-7} \text{ m}$
 $\nu = \frac{c}{\lambda} = \frac{3 \times 10^8}{6.2 \times 10^{-7}} = \frac{1}{2.04} \times 10^{15}$
 $= 0.49 \times 10^{15}$
 $= 4.9 \times 10^{14} \text{ s}^{-1}$

PHOTOELECTRIC EFFECT :->

→ In 1887, Hertz performed a very interesting experiment in which e^- s were ejected when certain metals (for eg $\rightarrow$ K, Rb, Cs etc) were exposed to a beam of light. and this phenomena is called Photoelectric effect.



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Observations :->

- ① The e^- s are ejected from the metal surface as soon as the beam of light strikes the surface. i.e. there is no time lag between the striking of light and the ejection of e^- s from the metal surface.
- ② The no. of e^- s ejected is proportional to the intensity or brightness of light.
- ③ For each metal there is a characteristic minimum frequency ν_0 (also known as threshold frequency) below which photoelectric effect is not observed. At a frequency $\nu > \nu_0$, the ejected electrons come out with certain kinetic energy. The KE of these e^- s increases with the increase of frequency of the light used.
- ④ The energy content of the beam of light depends upon the brightness of the light. In other words no. of electrons ejected and KE associated with them should depend on the brightness of light. It has been observed that though the no. of e^- s ejected does depend upon the brightness of light, the KE of the ejected e^- s does not.
- ⑤ KE of the ejected e^- s is proportional to the frequency of EM radiation. Since the striking photon has energy $= h\nu$ and the min energy required to eject the e^- is $h\nu_0$ (also called work function; ϕ_0). Then the difference in energy is $h\nu - h\nu_0$ is transferred as the KE of the photo e^- .

→ In photoelectric effect we have to consider two parameters. —

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- ① Intensity ② Frequency.

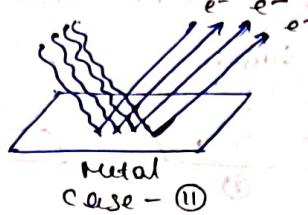
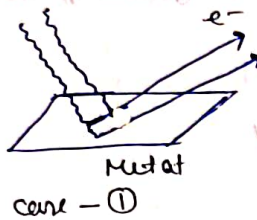
1. INTENSITY :→ (Brightness) Each photon have energy $[h\nu > h\nu_0]$

1 photon can eject only one electron

No. of photon $\propto$ brightness of light

$\propto$ no. of electron ejected

$\propto$ Photocurrent



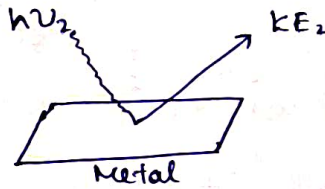
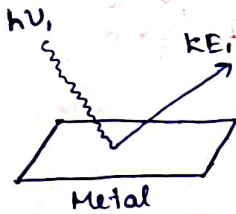
Photocurrent = ② > ① Photocurrent $\propto$ brightness

2. FREQUENCY :→ $[\nu > \nu_0]$

$$E \propto \nu$$

$$E = h\nu$$

as we, $\nu_T = R.F.T.$



$$\nu_2 \gg \nu_1$$

$$KE_2 \gg KE_1$$

MATHEMATICAL REPRESENTATION

OF PEE :→

$$h\nu = h\nu_0 + KE$$

$h\nu \Rightarrow$ incident radiation

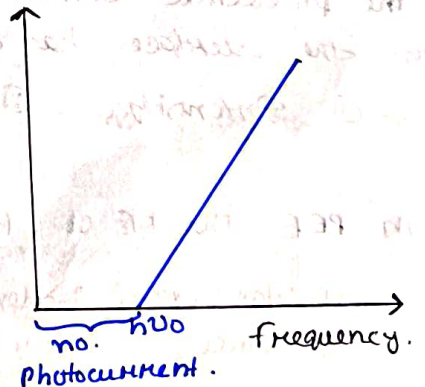
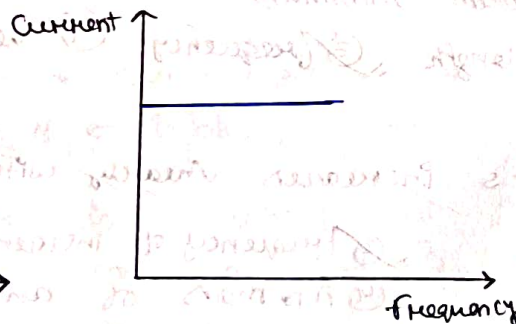
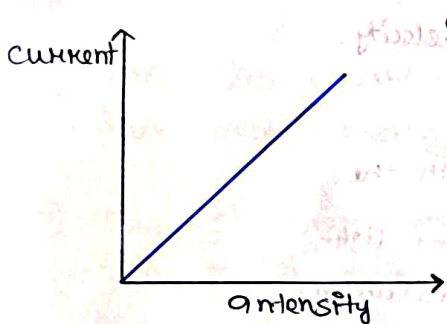
$h\nu_0 \Rightarrow$ Threshold frequency

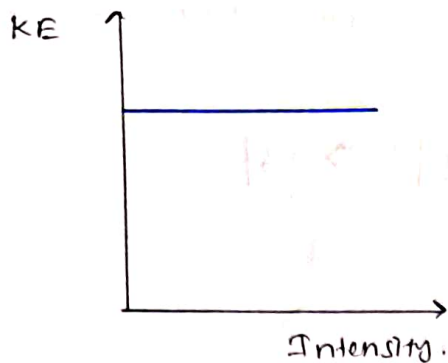
$KE \Rightarrow$ kinetic energy.

$$KE = h\nu - h\nu_0$$

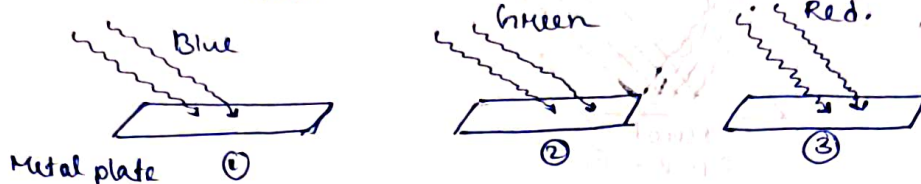
$$\frac{1}{2}mv^2 = h\nu - h\nu_0$$

GRAPHS RELATED TO PEE :→





Q Predict the probability to gain photo current for the following observations.



Soln: Photo current $\Rightarrow$ ① > ② > ③
 $\sim \Rightarrow$ ① > ② > ③
 Energy of incident photon = ① > ② > ③

Q A photon of wavelength $4 \times 10^{-7} \text{ m}$ strikes a metal surface the work function of the metal being 2.13 eV . The KE of emission would be,

- ① 0.97 eV ② 97 eV ③ $4.97 \times 10^{-19} \text{ eV}$ ④ $5.84 \times 10^5 \text{ eV}$

Soln: $\lambda = 4 \times 10^{-7} \text{ m} = 4 \times 10^3 \text{ \AA}$

$$\begin{aligned} KE &= \frac{hc}{\lambda} - 2.13 \\ &= \frac{12400}{4 \times 10^3} - 2.13 \\ &= 3.1 - 2.13 \\ &= 0.97 \end{aligned}$$

Q The photoelectric emission from a surface starts only when the light incident upon the surface has a certain minimum.

- ① Intensity ② wave length ③ frequency ④ Velocity.

Q In PEE the KE of photo e-s increases linearly with the -

- ① wavelength of incident light ② frequency of incident light
 ③ velocity " " ④ Atoms of an element.

BOHR'S ATOMIC MODEL :->

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ASSUMPTIONS :-

- ① Atom has a nucleus where all protons and neutrons are present. The size of nucleus is very small and it is present at the centre of atom.
- ② -vely charged e^- revolved around the nucleus in the same way as the planets revolve around the sun.

The path of e^- is circular and it is known as orbit. e^- s can only revolve around the nucleus in this orbit.

- ③ Every orbit has fixed amount of energy and the e^- s revolving in these orbits will have a fixed velocity.

④ QUANTISATION CONDITION

- ⑤ Every orbit has a fixed value of angular momentum of e^- which is the integral multiple of $\frac{h}{2\pi}$

$$L = n \frac{h}{2\pi}$$

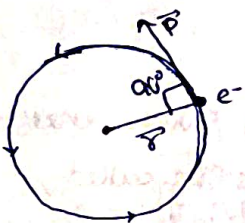
h = Planck's const.
 n = const.
 n = Orbit no. (integer)
 $n = 1, 2, 3, 4, \dots$

$$n=1, L = \frac{h}{2\pi}$$

$$n=2, L = \frac{2h}{2\pi}$$

$$n=3, L = \frac{3h}{2\pi}$$

The value of angular momentum can never be $\frac{1.5h}{2\pi}, \frac{2.5h}{2\pi}, \frac{3.5h}{2\pi}$ etc.



$$L = \vec{r} \cdot \vec{p}$$

$$L = \vec{r} \cdot m\vec{v}$$

$$L = r m v \sin \theta$$

$$L = m v r \sin 90^\circ$$

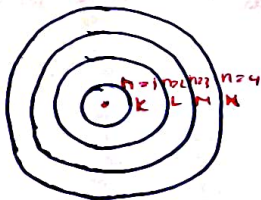
$$L = m v r$$

from ① and ②,

$$m v r = \frac{nh}{2\pi}$$

- ⑥ The orbits in which e^- can revolve are known as stationary orbits because in these orbits energy of e^- is always const.

- ⑦ Each stationary orbit is associated with definite amount of energy therefore these orbits are also called as energy levels and are numbered as K, L, M, N etc. from the nucleus outwards.

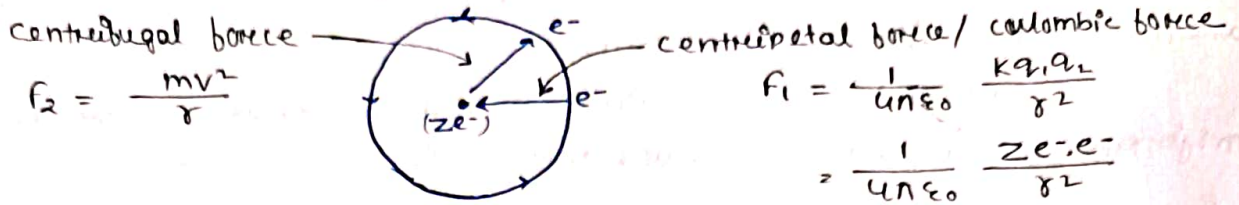


⑧ During the circular motion every e^- have two types of forces occurring on them

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① Centrifugal force : which operates opposite from nucleus.

② Centripetal force : which operates towards the nucleus and this is OR Coulombic force - the binding force.



At to the condition,

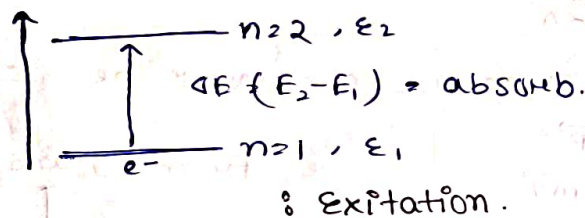
$$F_1 = F_2$$

$$\frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r^2} = \frac{mv^2}{r}$$

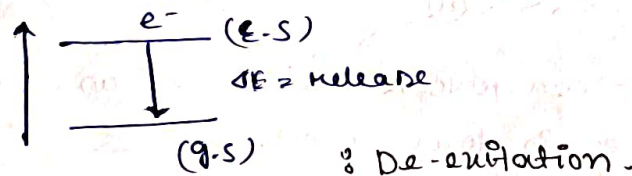
⑨ During the rotation e^- can never absorb or emit energy itself.

⑩ e^- can absorb ^{or emit} a particular amount of energy which is the exact difference of energy b/w two energy levels.

$\Rightarrow$ e^- jumps from lower energy level to higher energy level by absorbing certain amount of energy then the process is called excitation.



$\Rightarrow$ e^- goes down from higher energy level to lower energy level by emitting certain amount of energy. then the process is called de-excitation.



CALCULATIONS :-

1. RADIUS :

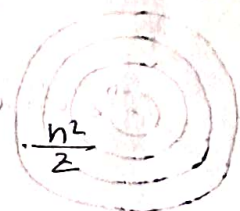
$$\frac{nh}{2\pi} = mvr \quad \text{--- (i)}$$

$$\frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r^2} = \frac{mv^2}{r} \quad \text{--- (ii)}$$

$$\text{(i)}^2 \div \text{(ii)} \Rightarrow \frac{n^2 h^2}{4\pi^2} \times \frac{4\pi\epsilon_0}{Ze^2} = \frac{m^2 v^2 r^2}{m v^2 r}$$

$$r = \frac{n^2 h^2 \epsilon_0}{\pi m Z e^2}$$

$$\Rightarrow r = \frac{h^2 \epsilon_0}{\pi m e^2} \cdot \frac{n^2}{Z}$$



$$r_n \Rightarrow 0.529 \times \frac{n^2}{Z} A$$

$$r_n = a_0 \frac{n^2}{Z} A$$

$$a_0 = 0.529 \approx 0.53$$

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Q Calculate the radius of 1st, 2nd, 3rd Bohr's orbit of Hydrogen.

Soln: - $n=1$

$$r = 0.529 \frac{1^2}{1} A$$

$$= 0.529 A$$

$n=2$

$$r = 0.529 \frac{2^2}{1} A$$

$$= 0.53 \times 4 A$$

$$= 2.12 A$$

$n=3$

$$r = 0.529 \frac{3^2}{1} A$$

$$= 0.53 \frac{9}{1} A$$

$$= 4.77 A$$

Q Calculate the radius ratio of 3rd and 5th orbit of He^+ .

Soln: - $r_3 = \frac{a_0 \frac{3^2}{2} A}{a_0 \frac{5^2}{2} A}$

$$= \frac{9}{25}$$

Q Calculate the radius ratio of 2nd orbit of H and 3rd orbit of Li^{2+} .

Soln: - $\frac{r_H}{r_{Li^{2+}}} = \frac{a_0 \frac{2^2}{1}}{a_0 \frac{3^2}{3}} = 4:3$

Q Calculate ratio of 2nd excited state of H and 6th excited state of Li^{2+} .

Soln: - $\frac{r_H}{r_{Li^{2+}}} = \frac{a_0 \frac{3^2}{1}}{a_0 \frac{2^2}{3}} = \frac{9 \times 3}{4} = 27:4$

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2. VELOCITY :

$$r = \frac{n^2 h^2 \epsilon_0}{\pi m Z e^2}$$

$$mvr = \frac{nh}{2\pi}$$

$$v = \frac{nh}{2\pi m r} = \frac{nh}{2\pi m} \times \frac{\pi m Z e^2}{n^2 h^2 \epsilon_0}$$

$$v_n = \frac{Ze^2}{2h\epsilon_0} \times \frac{Z}{n}$$

$$v_n = 2.18 \times 10^6 \left(\frac{Z}{n} \right) \text{ m/s}$$

$$\text{no. of revolutions per sec} = \frac{v_n}{2\pi r_n}$$

Q Calculate velocity of e^- placed in the 3rd orbit of Hydrogen. also calculate no. of revolutions per sec that this e^- makes around the nucleus.

Soln: - $v = 2.18 \times 10^6 \frac{1}{3}$

$$= 0.72 \times 10^6$$

$$= 7.2 \times 10^5$$

$$r = 0.53 \times \frac{3^2}{1}$$

$$= 4.77 A$$

$$= 4.77 \times 10^{-10} \text{ m}$$

$$\text{no. of revolutions} = \frac{7.2 \times 10^5}{2 \times 3.14 \times 4.77 \times 10^{-10}}$$

$$= 2.43 \times 10^{14} \text{ RPS}$$

3. ENERGY :

Total energy = KE + PE
(E)

$$v = \frac{ze^2 h}{2\pi n h \epsilon_0}, \quad r = \frac{n^2 h^2 \epsilon_0}{\pi m z e^2}$$

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$$\begin{aligned} \bullet \text{ KE} &= \frac{1}{2} m v^2 \\ &= \frac{1}{2} m \left(\frac{ze^2}{2\pi n h \epsilon_0} \right)^2 \\ &= \frac{1}{2} m \frac{z^2 e^4}{4 n^2 h^2 \epsilon_0^2} = \frac{1}{8} \frac{m e^4}{n^2 h^2 \epsilon_0^2} \\ &= \frac{1}{8} \frac{m e^4}{h^2 \epsilon_0^2} \times \frac{22}{n^2} \end{aligned}$$

$$\text{KE} \Rightarrow 13.6 \times \frac{z^2}{n^2} \text{ eV}$$

$$\begin{aligned} \bullet \text{ PE} &= \frac{k q_1 q_2}{r} = \frac{1}{4\pi \epsilon_0} \frac{ze(-e)}{r} = \frac{1}{4\pi \epsilon_0} \times \frac{\pi m z e^2}{n^2 h^2 \epsilon_0^2} \\ &= \frac{1}{4} \frac{z^2 e^4 m}{n^2 h^2 \epsilon_0^2} \end{aligned}$$

$$\text{PE} = \frac{2}{2} \times \frac{1}{4} \frac{z^2 e^4 m}{n^2 h^2 \epsilon_0^2} = -2 \left(\frac{1}{8} \frac{z^2 e^4 m}{n^2 h^2 \epsilon_0^2} \right)$$

$$\text{PE} = -2\text{KE}$$

$$\text{Total energy (E)} = \text{KE} + \text{PE} = \text{KE} - 2\text{KE} = -\text{KE}$$

$$E = -\text{KE}$$

$$\text{Energy of } e^- \text{ in any state} = \frac{E}{n^2}$$

$$E_n = -13.6 \frac{z^2}{n^2} \text{ eV}$$

IE = Ionisation Energy.

Q Calculate the energy of e^- in 1st, 2nd, 3rd and 4th orbit of H-atom

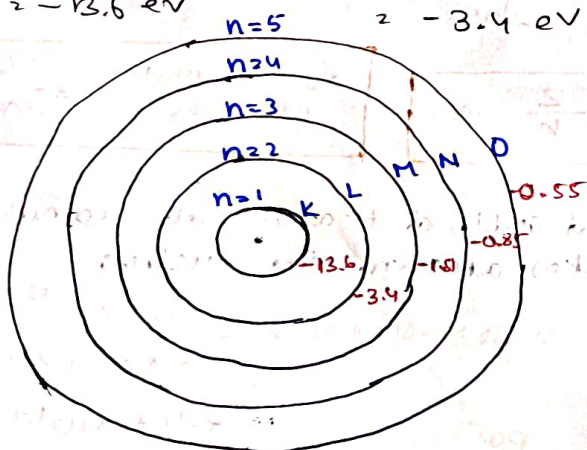
Soln:-

$$\begin{aligned} E_1 &= -13.6 \times \frac{1^2}{1^2} \\ &= -13.6 \text{ eV} \end{aligned}$$

$$\begin{aligned} E_2 &= -13.6 \times \frac{1^2}{2^2} \\ &= -13.6 \times \frac{1}{4} \\ &= -3.4 \text{ eV} \end{aligned}$$

$$\begin{aligned} E_3 &= -13.6 \times \frac{1^2}{3^2} \\ &= -13.6 \times \frac{1}{9} \\ &= -1.51 \text{ eV} \end{aligned}$$

$$\begin{aligned} E_4 &= -13.6 \times \frac{1^2}{4^2} \\ &= -13.6 \times \frac{1}{16} \\ &= -0.85 \text{ eV} \end{aligned}$$



NOTE:-

This formula is applicable for H-atom and H-like species i.e. single e^- species.

Since 'n' can have only integral values, it follows that total energy of the e^- is quantized.

The -ve sign indicates that the e^- is bound towards nucleus.

SOME EXTRA POINT :-

a) $KE = \frac{KZ^2}{2r}$, i.e., $KE \propto \frac{1}{r}$

$[r \uparrow KE \downarrow PE \uparrow TE \uparrow]$

b) $PE = -\frac{KZ^2}{r}$, i.e., $PE \propto -\frac{1}{r}$

$PE \propto r$

$[r \downarrow KE \uparrow PE \downarrow TE \downarrow]$

c) $TE = \frac{KZ^2}{2r}$, i.e., $TE \propto \frac{1}{r}$

$TE \propto r$

THE DIFFERENCE IN ENERGY B/W TWO ENERGY LEVELS :-

$$\Delta E = E_2 - E_1$$

$$= -13.6 \frac{Z^2}{n_2^2} - \left(-13.6 \frac{Z^2}{n_1^2} \right)$$

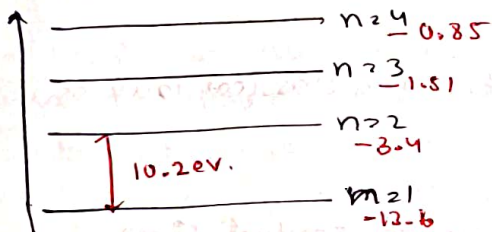
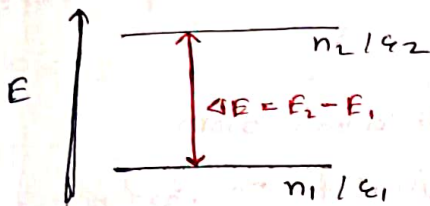
$$= 13.6 Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

Eg $\rightarrow n_1 = 1, n_2 = 2$ for H-atom.

$$\Delta E / \text{Excitation Energy} = 13.6 \times \left(\frac{1}{1} - \frac{1}{4} \right)$$

$$= 13.6 \times 1 - \frac{1}{4}$$

$$= 13.6 \times \frac{3}{4} = 10.2 \text{ eV.}$$



SOME IMPORTANT DEFINITIONS :-

1. **IONISATION ENERGY** : Minimum amount of energy required to liberate an e^- from the ground state of an isolated atom is called as I.E.

$$[n_1 = 1, n_2 = \text{infinite}]$$

2. **SEPARATION ENERGY** : Minimum energy required to remove an e^- from its excited state is called as separation energy.

$$[n_1 = 2, 3, 4, \dots, n_2 = \infty]$$

3. **EXCITATION ENERGY** : Amount of energy required to shift an e^- from ground state to any excited state is called as excitation energy.

$$[n_1 = 1, n_2 = 2, 3, 4, \dots]$$

o All these kinds of energy are always positive.

Q 96 the total energy of an e^- in -1.51 eV in H-atom then find out 108
 ① KE ② PE ③ ORBIT(n) ④ ORBITAL radius ⑤ velocity of e^- in that orbit

Soln:-
 ③ $-1.51 = -13.6 \frac{Z^2}{n^2}$
 $-1.51 = -13.6 \frac{1}{n^2}$
 $n^2 = \frac{13.6}{1.51} = 9$
 $n = 3$

① $TE = -KE$
 $-1.51 = -KE$
 $KE = 1.51$ eV

② $PE = -2KE$
 $= -3.02$ eV

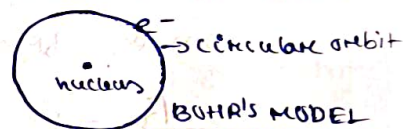
④ $r = 0.53 \times \frac{Z^2}{n^2}$
 $= 0.53 \times 9 = 4.77 \text{ \AA}$

⑤ $v = 2.18 \times 10^6 \frac{1}{n}$
 $= 0.779 \times 10^6$
 $= 7.2 \times 10^5$

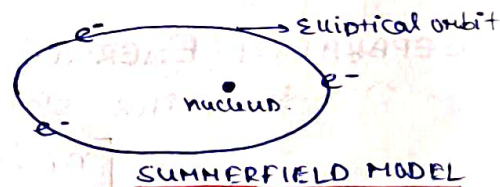
LIMITATIONS OF BOHR'S MODEL :-

- ① Bohr's theory does not explain the spectrum of multielectron atom.
- ② why the angular momentum of revolving e^- is equal to $\frac{nh}{2\pi}$, has not been explained by Bohr's theory.
- ③ Bohr inter related quantum theory of radiation and classical laws of physics without any theoretical explanation.
- ④ Bohr's theory does not explain the fine structure of spectral lines.
- ⑤ Bohr's theory does not obey Heisenberg uncertainty principle.

A/c to Bohr the orbits are circular so we can easily determine the position and momentum of any revolving e^-



A/c to Sommerfeld model the orbits are not circular but elliptical, so the positions and momentum of e^- are changing continuously. so we cannot calculate both of them simultaneously which follows Heisenberg's uncertainty principle.



[When $n = \infty$, classical mechanics merge with quantum mechanics.]

2/09/24

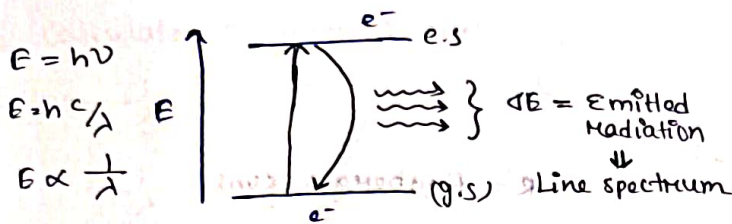
L-12

HYDROGEN SPECTRA :-

- When a radiation is passed through a spectroscope (prism) for the dispersion of the radiation, the pattern (photograph) obtained on the screen (photographic plate) is called the spectrum in the given radiation.

CLASSIFICATION :-

- Spectrum can be classified into two types: ① Emission ② Absorption.



V I B U V R

 $\lambda \uparrow$

- On heating e^- absorbs energy and jumps to higher energy level and it becomes unstable.

so e^- falls back or deexcites to a lower energy level by emitting energy in the form of radiation energy.

of such photon and e^- is equal to the difference in energy b/w two energy levels and generally they are different from each other for every e^- . The radiations will give particular lines at particular frequency and all the lines combined to give atomic spectra.

DEFINITION OF HYDROGEN SPECTRUM :-

- When an electric excitation is applied on hydrogen atomic gas at low pressure a bluish light is emitted. When a ray of this light is passed through a prism a spectrum of several isolated sharp lines is obtained. The wavelength of various lines show that spectrum lines lie in visible, UV, IR region. These lines are grouped into different series.

LYMEN SERIES :-

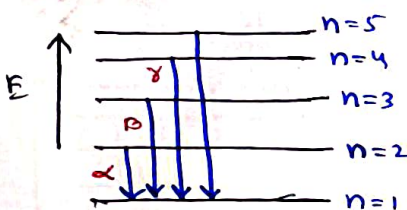
UV region (912 Å - 1216 Å)

 $\Rightarrow n_1 = 1, n_2 = 2, 3, 4, 5$

1st line / initial line /

starting line / α -line

Lowest energy line

 $E \propto \frac{1}{\lambda}$ lowest ν line $E \propto \frac{1}{\lambda}$ highest λ line.
 $\Rightarrow n_1 = 1, n_2 = \infty$

Last line / Terminal line

Highest energy transition

" " "

Lowest λ "

Q Calculate the energy and wavelength of first line of Lyman series.

Soln:- $E = 13.6 Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

$= 13.6 \left(1 - \frac{1}{4} \right) = 13.6 \times \frac{3}{4} = 3.4 \times 3 = 10.2 \text{ eV}$

$\lambda = \frac{12400}{10.2} \approx 1240 \text{ \AA}$

Rydberg's Const. $\rightarrow$

$\Delta E = E_2 - E_1$

$\Delta E = 13.6 \times Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

$\frac{hc}{\lambda} = \frac{13.6 \times hc}{hc} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

$\frac{1}{\lambda} = \frac{13.6}{hc} \left[\frac{n_2^2 - n_1^2}{n_1^2 \cdot n_2^2} \right]$

$\frac{1}{\lambda} = R \left[\frac{n_2^2 - n_1^2}{n_1^2 n_2^2} \right]$

$\lambda = \frac{1}{R} \left[\frac{n_1^2 n_2^2}{n_2^2 - n_1^2} \right]$

$R =$ Rydberg's const.

$\frac{1}{R} = 912 \text{ \AA}$

$R = 109678 \text{ cm}^{-1}$

$R = 109700 \text{ cm}^{-1}$

$= 10970000 \text{ m}^{-1}$

$= 1.1 \times 10^7 \text{ m}^{-1}$

$4.9 \cdot 24$
 $L - 13$

Q Calculate the wavelength of last line or limiting line of Lyman series

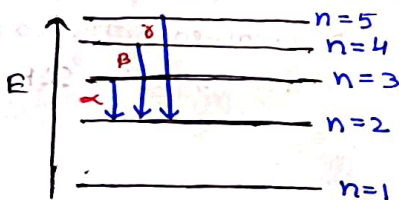
Soln:- $n_1 = 1, n_2 = \infty$

$\frac{1}{\lambda} = R Z^2 \left(\frac{1}{n_1^2} - \frac{1}{\infty^2} \right) = R [1]$

$\lambda = \frac{1}{R} = 912 \text{ \AA}$

BALMER SERIES $\rightarrow$ visible region (4000\AA - 7000\AA)

$n_1 = 2, n_2 = 3, 4, 5, \dots$



* Calculate the λ of first excited / α -line / min energy line / min frequency line / Maxm λ line.

Soln:- $n_1 = 2, n_2 = 3$

$\frac{1}{\lambda} = R Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

$\frac{1}{\lambda} = R \cdot 1 \cdot \frac{5}{36}$

$\lambda = \frac{1}{R} \cdot \frac{36}{5} = 912 \text{ \AA} \times \frac{36}{5} = 6566 \text{ \AA}$

Q Calculate the wavelength last of Balmer series

Soln:-

$$n_1 = 2, \quad n_2 = \infty$$

$$\frac{1}{\lambda} = R Z^2 \left(\frac{1}{4} - \frac{1}{\infty} \right)$$

$$\frac{1}{\lambda} = R \cdot \frac{1}{4}$$

$$\lambda = \frac{1}{R} \times 4 = 912 \text{ \AA} \times 4 = 3648 \text{ \AA}$$

Q Calculate the wavelength of 3rd line Bracket series in H spectrum.

Soln:- $\frac{1}{\lambda} = R Z^2 \left(\frac{1}{16} - \frac{1}{49} \right)$

$$n_1 = 4, \quad n_2 = 7$$

$$\frac{1}{\lambda} = R \left(\frac{49-16}{16 \times 49} \right) = R \frac{33}{766}$$

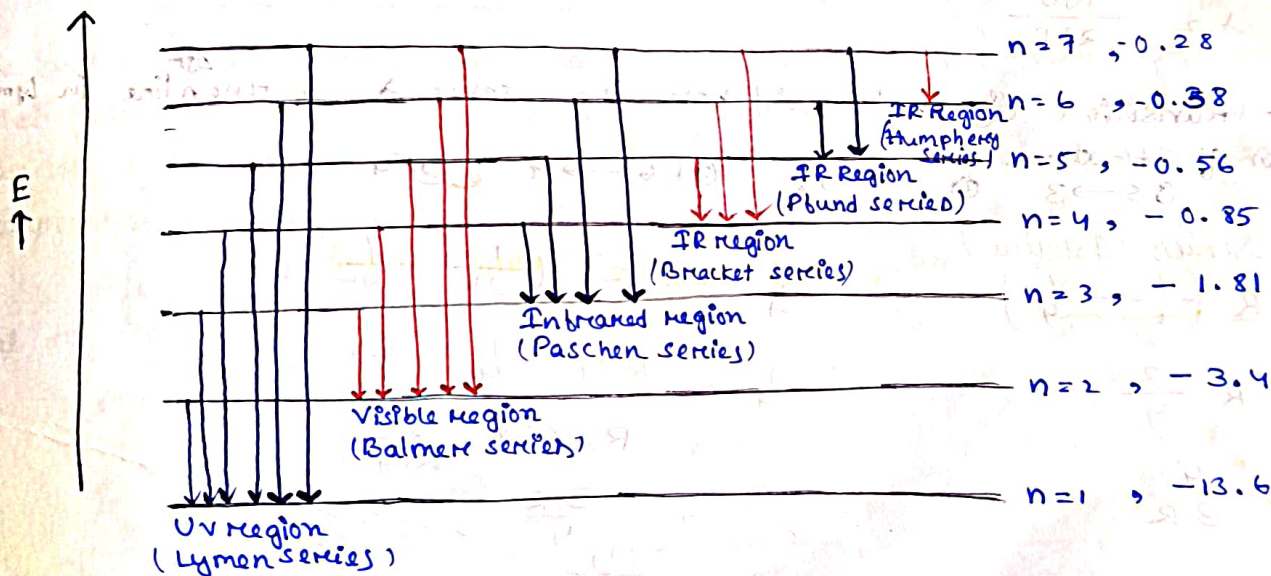
$$\lambda = \frac{1}{R} \times \frac{766}{33} = \frac{304}{912 \text{ \AA}} \times \frac{766}{33} = 21667 \text{ \AA}$$

ratio of

Q Calculate the wavelength of Lyman and Balmer series.

Soln:-

$$\frac{\lambda_L}{\lambda_B} = \frac{1240}{6566} = \frac{1}{5}$$

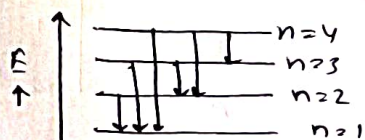


CALCULATION OF NO. OF SPECTRAL LINES :-

Q Calculate the total no. of spectral line from the following transition.

① $n_1 = 1$ to $n_2 = 4$.

$$\text{No. of spectral lines} = \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$$



no. of spectral lines = 6

$$= \frac{(4-1)(4-1+1)}{2} = \frac{12}{2} = 6$$

Q In the H-spectrum if the e^- moves from 7 to 1 by multistep transition then find out the total no. of lines.

Soln:- no. of spectral lines = $\frac{(7-1)(7-1+1)}{2} = \frac{42}{2} = 21$

Q In a H-spectrum e^- moves from 6th to 2nd orbit by multistep transition then find out the no. of lines.

Soln:- no. of spectral lines = $\frac{(6-2)(6-2+1)}{2} = \frac{4 \times 5}{2} = 10$

Q In which of the following transition the wavelength is min.

- ① $n=6$ to $n=4$ ② $n=4$ to $n=2$ ③ $n=3$ to $n=1$ ④ $n=2$ to $n=1$
more energy gap
 $\lambda \downarrow$

Q The wavelength of third line of Balmer series for a Hydrogen atom is

- ① $\frac{21}{100R}$ ② $\frac{100}{21R}$ ③ $\frac{21R}{100}$ ④ $\frac{100R}{21}$

Soln:- $n_1=2, n_2=5$

$$\frac{1}{\lambda} = R Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) = R \frac{25-4}{100} = \frac{21R}{100}$$

$$\lambda = \frac{100}{21R}$$

Q What transition of He^+ will have the same λ as the 1st line in Lyman series of H-atom.

- ① $5 \rightarrow 3$ ② $3 \rightarrow 2$ ③ $6 \rightarrow 4$ ④ $4 \rightarrow 2$

Soln:- Lyman series (1st line)

$$\frac{1}{\lambda} = R \left(\frac{1}{1^2} - \frac{1}{4^2} \right)$$

$$\frac{1}{\lambda} = R \frac{3}{4}$$

$$\lambda = \frac{4}{3R}$$

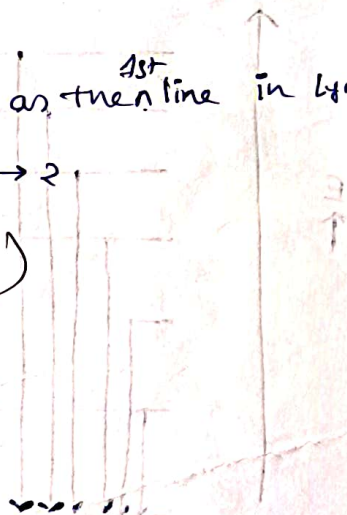
$$He^+ \rightarrow \frac{1}{\lambda} = R Z^2 \left(\frac{1}{2^2} - \frac{1}{4^2} \right)$$

$$\frac{1}{\lambda} = R \times 4 \left(\frac{1}{4} - \frac{1}{16} \right)$$

$$= R \times 4 \times \frac{3}{16}$$

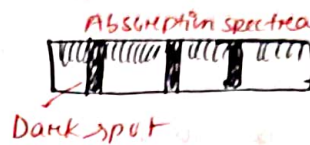
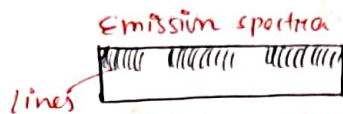
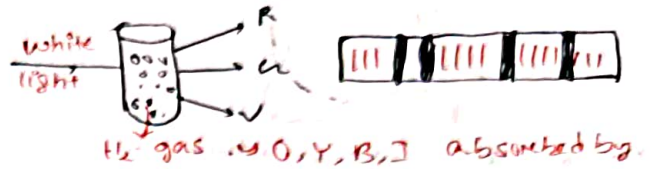
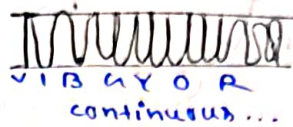
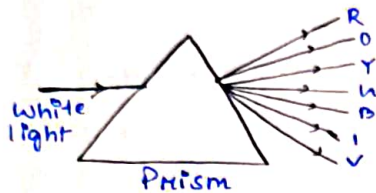
$$= \frac{3R}{4}$$

$$\lambda = \frac{4}{3R}$$



HYDROGEN ABSORPTION SPECTRA :-

- An absorption spectrum is a type of spectrum of light that contains a pattern of dark lines or bands and these lines are created when e^- absorbs energy to reach a higher energy state.



WAVE PARTICLE DUALITY :-

- This model consist of the following —
- ① Debroglie's Hypothesis
 - ② Heisenberg's Uncertainty principle

THE WAVE NATURE OF ELECTRON / PARTICLE

OR DEBROWLIE'S HYPOTHESIS :-

- The wave nature of light rays and x-rays is proved on the basis of their interference and diffraction and many facts related to radiations can only be explained when the beam of light rays is regarded as composed of energy packet or photons whose velocity is 3×10^{10} cm/sec.

A/c to Debroglie the wavelength, λ of an e^- is inversely proportional to its momentum P .

$$E = mc^2$$

$$E = h\nu = \frac{hc}{\lambda}$$

$$mc^2 = \frac{hc}{\lambda}$$

$$\Rightarrow m \cdot c \cdot \lambda = \frac{hc}{c}$$

$$\lambda = \frac{h}{mc}$$

$$\lambda = \frac{h}{p}$$

$$\lambda \propto \frac{1}{m}$$

- It is clear from the above equation that the value λ decreases with increasing either m or v or both.

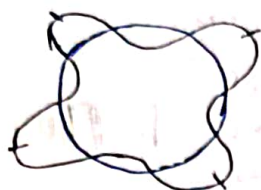
The wavelength, many body moving objects like an aeroplane, or a cricket ball, is very low because of their high mass.

$$v \propto \frac{1}{\lambda}$$

BOHR'S THEORY AND DEBROGLIE'S CONCEPT: →

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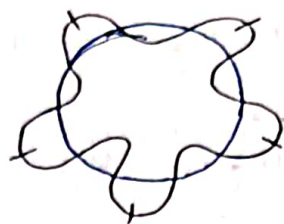
- ① Acc to Debroglie the nature of an e- moving around the nucleus is like a wave that flows in circular orbit around the nucleus.



no. of orbit = no. of waves

orbit = 4

wave = 4



$n=5$

wave = 5.

- ② As an e- is regarded as a wave the quantum condition as given by Bohr in his theory is readily fulfilled.
- ③ As the radius of circular orbit is r , then its circumference is $2\pi r$
- ④ From Bohr's theory the angular momentum,

$$mv r = \frac{nh}{2\pi}$$

$$2\pi r = \frac{nh}{mv} = \frac{nh}{p} = n\lambda$$

Q. In H-atom if r is the radius find Bohr orbit de Broglie wavelength of an e- in 3rd orbit is - (1) $3\pi r$ (2) $6\pi r$ (3) $\frac{3r}{2}$ (4) $\frac{r}{2}$

Soln:- $r_1 = 0.529 \frac{h^2}{2m} = 0.529 \frac{h^2}{2m} = 0.529 r_1$ $r_3 = 0.529 \times 9 = 4.761 r_1 = 4.761 \times 0.529 \text{ Å} = 2.52 \text{ Å}$

$$2\pi r = n\lambda$$

$$\lambda = \frac{2\pi r}{n} = \frac{2\pi \times 4.761 r_1}{3} = 6\pi r_1$$

DEBROGLIE'S WAVELENGTH IN TERMS OF KE: →

$$KE = \frac{1}{2} m v^2$$

$$m KE = \frac{1}{2} m^2 v^2$$

$$m^2 v^2 = 2 m KE$$

$$mv = \sqrt{2 m KE}$$

$$\lambda = \frac{h}{mv}$$

$$\lambda = \frac{h}{\sqrt{2 m KE}}$$

$$\lambda = \frac{h}{\sqrt{2 m eV}}$$

When a charged particle carrying q coulomb charge is accelerated by applying potential difference of V volt then

$$KE = eV$$

Q The mass of a particle is 1mg and its velocity is $4.5 \times 10^5 \text{ m/s}$ what would be the wavelength of this wavelength, if $h = 6.6 \times 10^{-27} \text{ erg sec}$

Soln:- $\lambda = \frac{h}{mv} = \frac{6.6 \times 10^{-27}}{10^{-3} \times 4.5 \times 10^5} = \frac{6.6}{4.5} \times 10^{-29} \text{ (1 erg = } 10^7 \text{ J)}$
 $= 1.4 \times 10^{-29}$

Q which of the following should be the wavelength of an electron if its mass is $9.1 \times 10^{-31} \text{ kg}$ and its velocity is $1/10$ that of light

Soln:- $\lambda = \frac{h}{mv} = \frac{6.6 \times 10^{-34}}{9.1 \times 10^{-31} \times \frac{3}{10} \times 10^8} = \frac{6.6}{9.1 \times 3} \times 10^{-11}$
 $= 0.24 \times 10^{-10} = 2.4 \times 10^{-11} \text{ m}$

Q A ball having mass of 25g moves with velocity of $6.6 \times 10^7 \text{ cm/s}$ then find out the de-Broglie's wavelength

Soln:- $\lambda = \frac{h}{mv} = \frac{6.6 \times 10^{-34}}{25 \times 6.6 \times 10^5}$
 $= 0.04 \times 10^{-39}$
 $= 4 \times 10^{-41}$

HEISENBERG'S UNCERTAINTY PRINCIPLE : →

In 1927, Werner Heisenberg presented a principle known as Heisenberg uncertainty principle which states that it is impossible to calculate both the position and momentum of a microscopic particle simultaneously with great accuracy. If the uncertainty position is Δx and uncertainty momentum is ΔP_x then A/C Heisenberg,

$$\Delta x \cdot \Delta P_x \geq \frac{h}{4\pi}$$

$$\Delta x \cdot m \Delta v_x \geq \frac{h}{4\pi}$$

$$\Delta v_x \cdot \Delta x \geq \frac{h}{4\pi m}$$

$\Delta x = \text{error in position}$
 $\Delta P_x = \text{error in momentum}$

from the above equation,

when $\Delta x = 0$, then $\Delta v_x = \infty$

when $\Delta v = 0$, $\Delta x = \infty$

So if the position known quite accurately, the i.e., Δx is very small; Δp_x become large and vice versa. 116

Q Calculate the uncertainty in the position of a particle when the uncertainty in momentum is - (a) 10^3 g cm/s^{-1} (b) 0.

Soln:- (a) $\Delta x \cdot \Delta p = \frac{h}{4\pi}$

$$\Delta x = \frac{6.6 \times 10^{-27} \text{ erg}}{4 \times 3.14 \times 10^{-3}}$$

$$= 0.5 \times 10^{-24} = 0.527 \times 10^{-24} \text{ cm}$$

Q The uncertainty in the position and velocity of a particle are 10^{-10} m and $5.27 \times 10^{-24} \text{ m/s}$ respectively. Calculate the mass of the particle. 0.099 kg.

Soln:- $\Delta x \cdot m \Delta v = \frac{h}{4\pi}$

$$m = \frac{6.6 \times 10^{-34}}{10^{-10} \times 5.27 \times 10^{-24} \times 4 \times 3.14}$$

$$= 0.099 \text{ kg}$$

Q Calculate the uncertainty in velocity of a cricket ball of mass 150g if the uncertainty in its position is of the order of 1 Å. $3.49 \times 10^{-24} \text{ m/s}$

Soln:- $\Delta x \cdot m \Delta v = \frac{h}{4\pi}$

$$\Delta v = \frac{6.6 \times 10^{-34}}{2.4 \times 3.14 \times 0.15 \times 10^{-10}}$$

$$= \frac{1}{0.3} \times 10^{-24} = 3.49 \times 10^{-24} \text{ m/s}$$

QUANTUM MECHANICAL MODEL OF ATOM :->

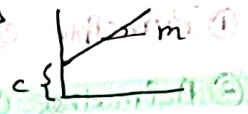
Quantum mechanics was developed independently in 1926 by Werner Heisenberg and Schrödinger.

The fundamental equation of quantum mechanics was developed by Schrödinger and this equation incorporates wave particle duality of matter as proposed by De-Broglie.

WAVE FUNCTION :->

- On quantum mechanics, a wave function is a mathematical description of a particle's quantum state.
- It is a function of variables like time, momentum, spin, position. The value of a wave function at a given point in space and time is

related to the likelihood of a particle being there at that time.
 wave function, $\Psi(x)$ → eg → $y = mx + c$
 $y = 2x + 4$
 ① Energy ② Position of e^- ③ Velocity.



SCHRÖDINGER WAVE EQUATION : →

• for a system such as an atom or a molecule whose energy does not change with time, the Schrödinger equation is written as,

$$H\Psi = E\Psi$$

H = Hamiltonian operator
 E = Energy, Eigen value.
 Ψ = Wave function

eg → $\left[\frac{d}{dx} \right] \Psi = 2\Psi$
 Operator
 wave function.

• This eqn define the nature of st. line similarly Ψ will define the nature of e^- .

• Schrödinger give a recipe of constructing this operator from the expression of total energy of system. The total energy of the system takes into account the KEs of all the sub-atomic particles (e-s, nuclei) attractive potential b/w e-s and nuclei and repulsive potential among the e-s and nuclei individually. Solution of the equation gives E and Ψ

• If we solve the Schrödinger wave equation, we will get -

$$\frac{d^2\Psi}{dx^2} + \frac{d^2\Psi}{dy^2} + \frac{d^2\Psi}{dz^2} + \frac{\hbar^2 m}{2} + (E - V)\Psi = 0$$

$\hbar$ = Planck's const.
 E = Total energy
 V = Potential energy
 Ψ = wave function
 m = mass.

PROBABILITY DENSITY : (Ψ^2) →

• The probability of finding an e^- at a point within an atom is proportional to the square of the orbital wave function i.e. Ψ^2 which is known as probability density, and is always positive. From the value of Ψ^2 at different points within an atom it is possible to predict the region around the nucleus where e^- will most probably be found.

QUANTUM NUMBERS :-

To obtain complete information about an e^- in an atom 4 identification no. are required and these identification no. are called quantum no.

- ① Principle Quantum number (n) $\rightarrow$ Country (Shell / Orbit no.)
- ② Azimuthal Quantum number (l) $\rightarrow$ State (Subshell $\rightarrow$ s, p, d, f)
- ③ Magnetic Quantum number (m_l) $\rightarrow$ District (Orbitals $\rightarrow$ kamra)
- ④ Spin Quantum number (m_s) $\rightarrow$ House no.

1. PRINCIPLE QUANTUM NUMBER (n) :-

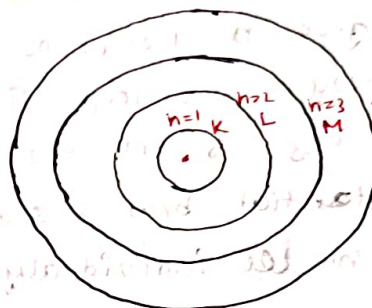
It represents the name and energy of the shell to which e^- belongs and size of orbital.

value of n varies from $1 \rightarrow \infty$

with increasing the value of n radius always increases.

$$r_n = 0.529 \times \frac{n^2}{Z} \text{ \AA}$$

$$r_1 < r_2 < r_3 < r_4 \dots$$



Greater the value of n greater is the energy of the shell.

$$E_n = -13.6 \frac{Z^2}{n^2} \quad E \propto \frac{1}{n^2}$$

$$E_1 < E_2 < E_3 < E_4 \dots$$

Greater the value of n , lesser the velocity.

$$v_n = 2.18 \times 10^6 \frac{Z}{n} \quad v_n \propto \frac{1}{n}$$

$$v_1 > v_2 > v_3 \dots$$

The angular momentum of revolving e^- is, $mvr = \frac{nh}{2\pi} \quad mvr \propto n$

De-broglie wavelength,

$$2\pi r = n\lambda \quad \lambda = \frac{2\pi r}{n} \quad \lambda \propto \frac{1}{n}$$

Maximum no. of e^- in a particular orbit = $2n^2$

Q Calculate no. of e^- for 7th period and 4th period.

Soln: no. of e^- (7) = 2×7^2
 $= 2 \times 49 = 98$

no. of e^- (4) = 2×4^2
 $= 2 \times 16$
 $= 32$

2. AZIMUTHAL / SECONDARY / ANGULAR QUANTUM NUMBER (l)

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• It represents the name of the sub-shell, shape of the orbital and orbital angular momentum.

• Value of 'l' $\Rightarrow 0 \rightarrow (n-1)$ eg $\rightarrow$ $n=1, l=0$
 $n=2, l=0, 1$
 $n=5, l=0, 1, 2, 3, 4$

RELATION B/W l & SUBSHELL :

l	0	1	2	3	4	5
Sub Shell	s	p	d	f	g	h

For diff. orbitals,

$n=1, l=0 \rightarrow 1s$

$n=3, l=0, 1, 2 \rightarrow 3s, 3p, 3d$

$n=2, l=0, 1 \rightarrow 2s, 2p$

$n=4, l=0, 1, 2, 3 \rightarrow 4s, 4p, 4d, 4f$

• The no. of e^- s in a particular subshell = $2(2l+1)$

for s-orbital,

$$l=0, \quad 2(2l+1) = 2(0+1) = 2$$

for p-orbital,

$$l=1, \quad 2(2l+1) = 2 \times 3 = 6$$

for d-orbital,

$$l=2, \quad 2(2l+1) = 2 \times 5 = 10$$

for f-orbital,

$$l=3, \quad 2(2l+1) = 2 \times 7 = 14$$

for g-orbital,

$$l=4, \quad 2(2l+1) = 2 \times 9 = 18$$

for h-orbital,

$$l=5, \quad 2(2l+1) = 2 \times 11 = 22$$

• If the value of n is same, then the order of energy of the various subshell will be $\rightarrow s < p < d < f$ eg $\rightarrow 4s < 4p < 4d < 4f$

• If the value of l is same, but value of n is different then the order of energy will be $\rightarrow$ greater 'n', greater energy $n \uparrow E \uparrow$

$$\text{eg} \rightarrow 1s < 2s < 3s < 4s$$

o Value of orbital angular momentum

$$\rightarrow L = \sqrt{l(l+1)} \frac{h}{2\pi}$$

$$= \sqrt{l(l+1)} \hbar \left[\hbar = \frac{h}{2\pi} \right]$$

eg $\rightarrow$ 2s $\quad$ 5s
 $\rightarrow L = 0 \quad L = 0$

3p
 $L = \sqrt{1(1+1)} \frac{h}{2\pi}$
 $= \frac{\sqrt{2}h}{2\pi}$
 $= \frac{h}{\sqrt{2}\pi}$

3d
 $L = \sqrt{2(2+1)} \frac{h}{2\pi}$
 $= \sqrt{6} \frac{h}{2\pi}$

5d
 $L = \sqrt{3(3+1)} \frac{h}{2\pi}$
 $= \sqrt{12} \frac{h}{2\pi}$

o With the value of l the shape of the orbital vary.

ORBITAL

SHAPE

s-orbital $\rightarrow$ Spherical 

p-orbital $\rightarrow$ Dumbbell 

d-orbital $\rightarrow$ Double dumbbell 

f-orbital $\rightarrow$ Complex.

3. MAGNETIC QUANTUM NUMBER (m_l):

o It represents the orientation of e^- cloud

Under the influence of magnetic field each subshell is further divided into orbitals where the e^- clouds are present. Magnetic quantum no. describes these different distributions of e^- cloud.

o The value of Magnetic quantum no varies from $-l \rightarrow +l$.

$\rightarrow n=1 \quad l=0 \quad m_l=0$

$\rightarrow n=2 \quad l=0 \quad m_l=0$
 $\quad \quad \quad 1 \quad \quad \quad +1$
 $\quad \quad \quad \quad \quad \quad 0$
 $\quad \quad \quad \quad \quad \quad -1$

1s

2s

2p
 $+1 \quad 0 \quad -1$
 $-1 \quad 0 \quad +1$

1 p-orbital = 2e-
 Total p-orbital = 6e-

$p_x \quad p_z \quad p_y$
 $p_y \quad p_z \quad p_x$

3s

3p
 $+1 \quad 0 \quad -1$
 $-1 \quad 0 \quad +1$

3d
 $+2 \quad +1 \quad 0 \quad -1 \quad -2$
 $-2 \quad -1 \quad 0 \quad +1 \quad +2$

$d_{xy} \quad d_{yz} \quad d_{z^2} \quad d_{xz} \quad d_{yz}$

$\rightarrow n=3 \quad l=0 \quad m_l=0$

$\quad \quad \quad 1 \quad \quad \quad +1$
 $\quad \quad \quad 2 \quad \quad \quad 0$
 $\quad \quad \quad \quad \quad \quad -1$

$+2$
 $+1$
 0
 -1
 -2

ORBITAL :->

3D space around the nucleus where the probability of finding e- is maximum is called an orbital. An orbital can be represented by 3-set of quantum no.s.

$$Eg \rightarrow Y_{210} \Rightarrow \left. \begin{matrix} n=2 \\ l=1 \\ m_l=0 \end{matrix} \right\} 2p_z$$

$$Y_{311} \Rightarrow \left. \begin{matrix} n=3 \\ l=1 \\ m_l=1 \end{matrix} \right\} 3p_x \text{ or } 3p_y$$

Q Find out the orbitals or representation of the following ->

$$① Y_{520} \Rightarrow \left. \begin{matrix} n=5 \\ l=2 \\ m_l=0 \end{matrix} \right\} 5p_z$$

$$② Y_{100} \Rightarrow \left. \begin{matrix} n=1 \\ l=0 \\ m_l=0 \end{matrix} \right\} 1s$$

$$③ Y_{32+1} \Rightarrow \left. \begin{matrix} n=3 \\ l=2 \\ m_l=+1 \end{matrix} \right\} 3d_x \text{ or } 3d_y$$

$$④ Y_{620} \Rightarrow \left. \begin{matrix} n=6 \\ l=2 \\ m_l=0 \end{matrix} \right\} 6d_{z^2}$$

$$⑤ 4p_x \rightarrow Y_{41+1}$$

$$⑥ 5f \rightarrow Y_{53}$$

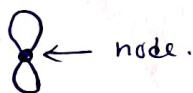
$$⑦ 6d_{z^2} \rightarrow Y_{620}$$

$$⑧ 2p_y \rightarrow Y_{21+1}$$

NODE :->

On which place in an atom the probability of finding angle is zero is called node.

There are two types of node ->



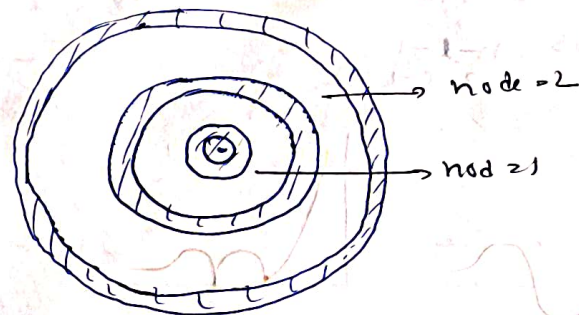
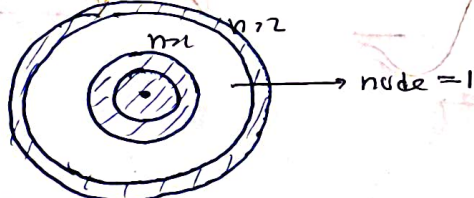
① Nodal surface or spherical nodes or Radial nodes.

$$\text{no. of Radial nodes} = n - l - 1$$

$$1s, n-l-1 = 1-0-1 = 0$$

$$3s, n-l-1 = 3-0-1 = 2$$

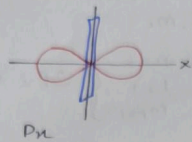
$$2s, n-l-1 = 2-0-1 = 1$$



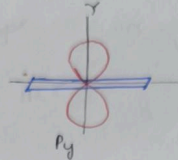
③ Angular node or nodal plane :

$$\text{no. of angular nodes} = l$$

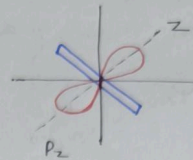
eg → p-orbital,



y-z plane



z-x plane

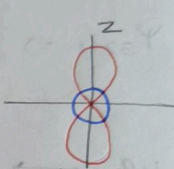
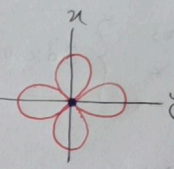
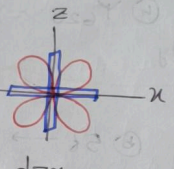
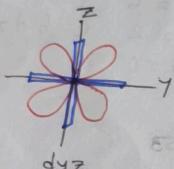
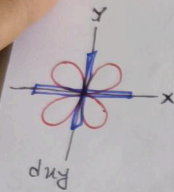


x-y plane

$$l = 1, \text{ Angular node} = 1$$

d-orbital,

$$l = 2, \text{ Angular node} = 2 \text{ (nodal plane)}$$



Set → 1 : in between the axis

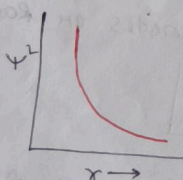
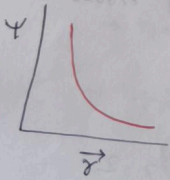
Set → 2 : Along the axis

$$\text{Total no. of nodes} = n - 1$$

RADIAL DISTRIBUTION :

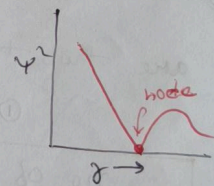
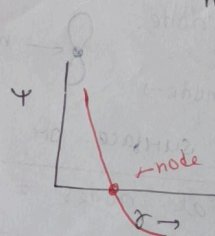
① 1s

$$\begin{aligned} \text{Radial node} &= n - l - 1 \\ &= 1 - 0 - 1 \\ &= 0 \end{aligned}$$



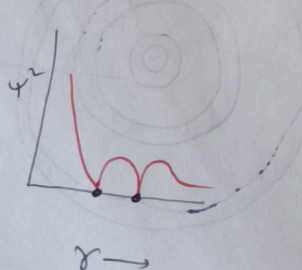
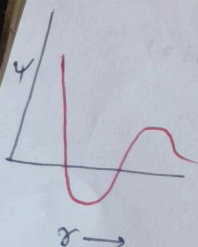
② 2s

$$n - l - 1 = 2 - 0 - 1 = 1$$



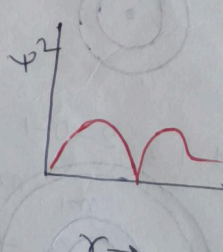
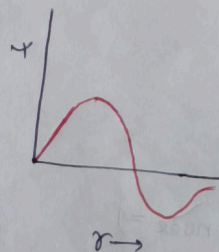
③ 3s

$$\begin{aligned} n - l - 1 &= 3 - 0 - 1 \\ &= 2 \end{aligned}$$



④ 3p

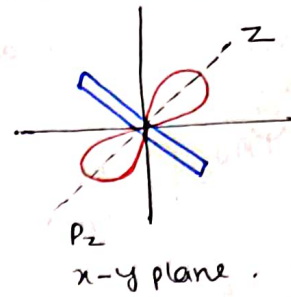
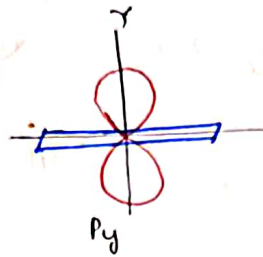
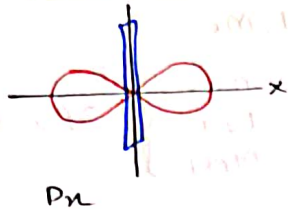
$$n - l - 1 = 3 - 1 - 1 = 1$$



② Angular node or nodal plane :

no. of angular nodes = l

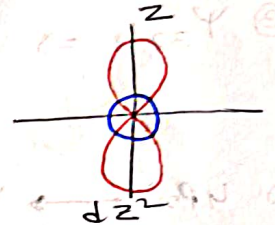
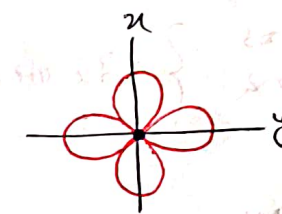
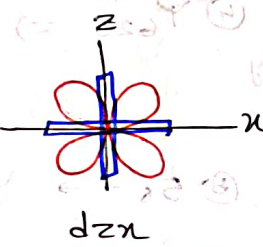
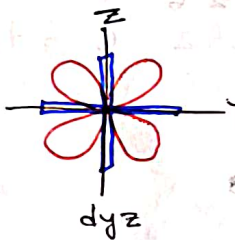
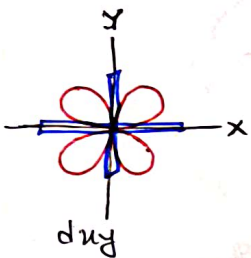
eg → p-orbital,



$l=1$, Angular node = 1

d-orbital,

$l=2$, Angular node = 2 (nodal plane)



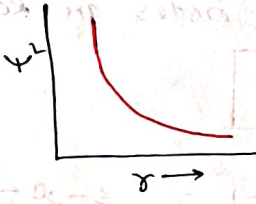
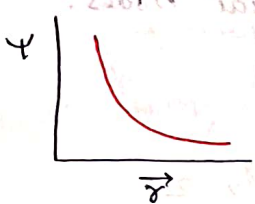
Set → 1 : in between the axis

Set → 2 : Along the axis

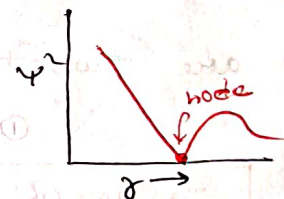
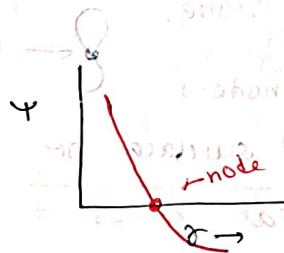
Total no. of nodes = $n-1$

RADIAL DISTRIBUTION :

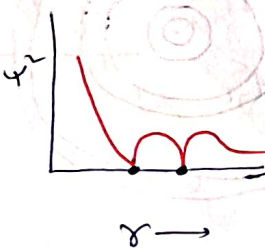
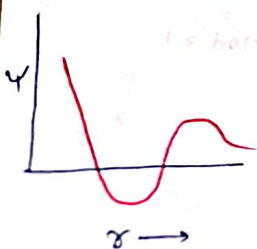
① 1s Radial node = $n-l-1$
= $1-0-1$
= 0



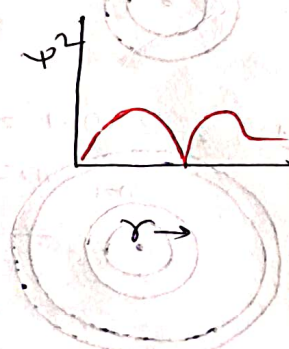
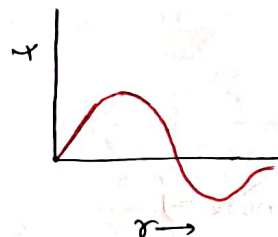
② 2s $n-l-1 = 2-0-1 = 1$



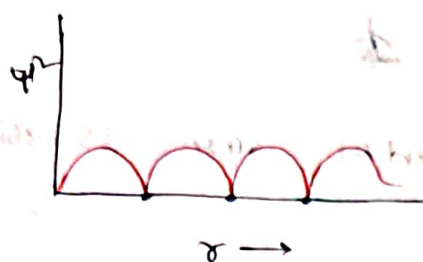
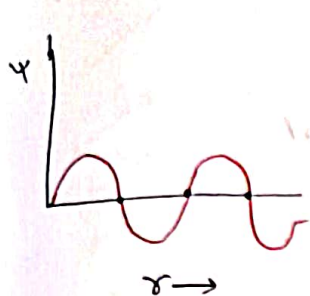
③ 3s $n-l-1 = 3-0-1 = 2$



④ 3p $n-l-1 = 3-1-1 = 1$



⑤ $60 \quad n-l-1 = 6-2-1 = 3$



$2\pi r = n\lambda$

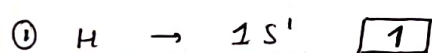
$\lambda = \frac{2\pi r}{n}$
 $= \frac{2\pi a n}{2}$

4. SPIN QUANTUM NUMBER (m_s) :

- It represents the direction of e^- spin around its axis.
- For CW spin if we consider the value of S $+\frac{1}{2}$ then for ACW spin the value will be $-\frac{1}{2}$ and vice versa.

Spin angular momentum $= \sqrt{S(S+1)} \frac{h}{2\pi} = \sqrt{S(S+1)} \hbar$ $\hbar = \frac{h}{2\pi}$

Q Calculate the spin angular momentum for - ① H, ② N



Spin contribute from unpaired e^- only.

no. of unpaired $e^- = 1$

for each $e^- = +\frac{1}{2}$

Spin angular momentum $= \sqrt{S(S+1)} \frac{h}{2\pi}$
 $= \sqrt{\frac{1}{2}(\frac{1}{2}+1)} (\frac{h}{2\pi})$
 $= \sqrt{\frac{1}{2} \cdot \frac{3}{2}} \frac{h}{2\pi}$
 $\Rightarrow \frac{\sqrt{3}}{2} \frac{h}{2\pi} = \frac{\sqrt{3}h}{4\pi}$



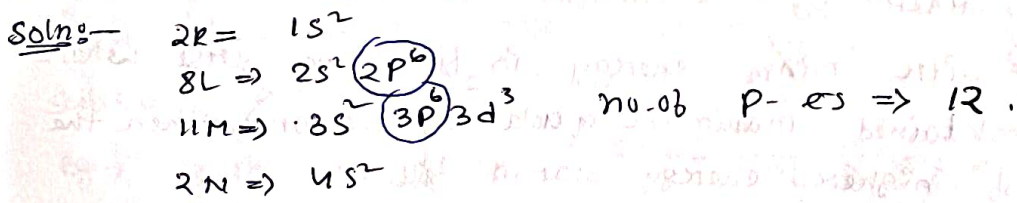
no. of unpaired $e^- = 3$
 for each $e^- =$

• Each orbital can accommodate 2 e^- with opposite spin or spin pair.

Q A neutral atom of an element has 2K, 8L, 11M and 2N e^- s.

The no. of p-electrons in the atom are—

- ① 2 ② 12 ③ 10 ④ 6.



K, $n=1$

L, $n=2$

M, $n=3$

N, $n=4$

Q Calculate the value of n, l , and m for $7p$ orbital.

Soln: $n = 7, l = 1, m = \pm 1$

Q Calculate the value of n, l and m for $3s$ orbital

Soln: $n = 3, l = 0, m = 0$

Q Which of the following set of quantum no.s is not possible.

- a) $n=3, l=0, m=-1, s=\pm\frac{1}{2}$ b) $n=3, l=2, m=0, s=\pm\frac{1}{2}$
 c) $n=2, l=3, m=-2, s=\pm\frac{1}{2}$

Q Calculate the total no. of e^- present in $n=1, 2, 3, 4$

Soln:-

$n=1, l=0, m=0$	$\boxed{\uparrow\downarrow}$ 1s
$n=2, l=0, m=0$	$\boxed{\uparrow\downarrow}$ 2s
$m = +1, 0, -1$	$\boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow}$ 2p
$n=3, l=0, m=0$	$\boxed{\uparrow\downarrow}$ 3s
$m = +1, 0, -1$	$\boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow}$ 3p
$m = +2, +1, 0, -1, -2$	$\boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow}$ 3d
$n=4, l=0, m=0$	$\boxed{\uparrow\downarrow}$ 4s
$m = +1, 0, -1$	$\boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow}$ 4p
$m = +2, +1, 0, -1, -2$	$\boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow}$ 4d
$m = +3, +2, +1, 0, -1, -2, -3$	$\boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow} \boxed{\uparrow\downarrow}$ 4f

no of $e^- = 2n^2$
 no of orbital = n^2
 no of subshell = $2(2l+1)$

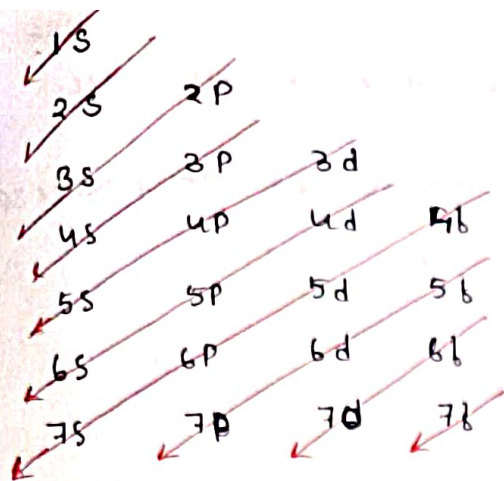
$(n+1)$ value ko hote energy 4 ko hote hote shell agote bill hote. Aru jodi $(n+1)$ same ho jai the prin. quan. no jithe ko hote hote agote bill hote.

RULES OF FILLING ELECTRONS :-

1. AUFBAU PRINCIPLE :-

→ Aufbau principle gives a sequence in which various sub-shell are filled up depending on the relative order of the energies of various shell.

Principle: The subshell with minm energy is filled up first when the subshell obtained maximum quota of electrons. then the next subshell of higher energy starts filling.

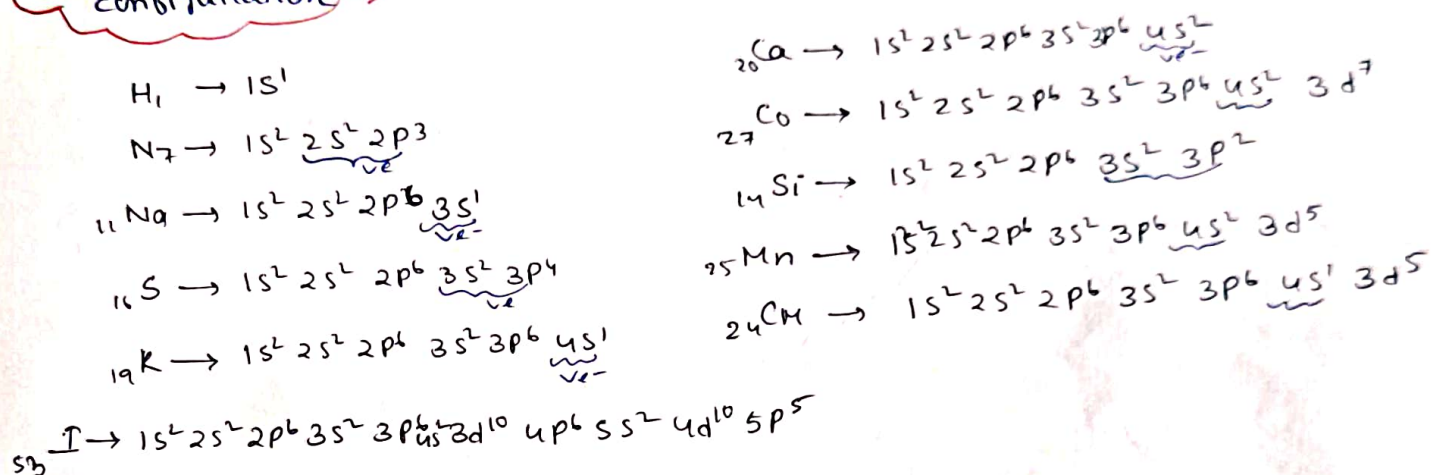


Energy order :

$$1s < 2s < 2p < 3s < 3p < 4s < 3d \\ < 4p < 5s < 4d < 5p \dots$$

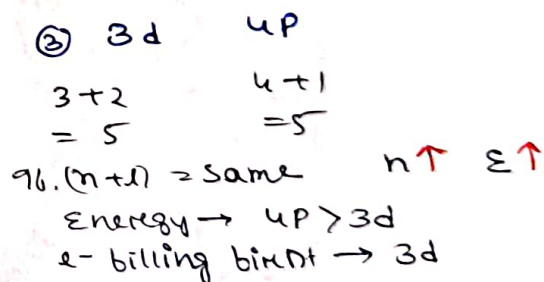
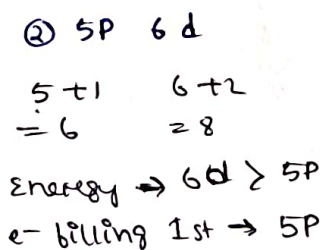
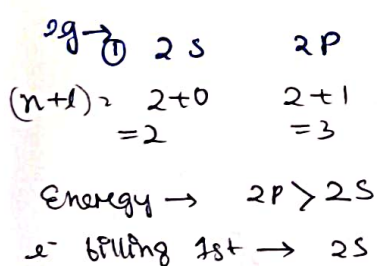
Penultimate shell $\rightarrow (n-1)$
 Pre " " $\rightarrow (n-2)$

Electronic configuration $\Rightarrow$



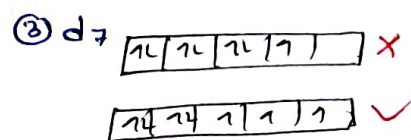
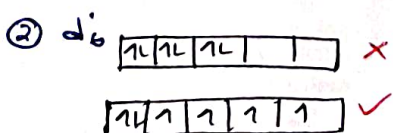
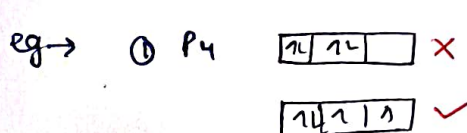
2. (n+l) RULE :

$\rightarrow$ The subshell with lowest (n+l) value are filled up first. When two or more subshell has same (n+l) value then the subshell with lowest value of n filled up first.



3. HUND'S RULE :

$\rightarrow$ Pairing of e^- s in the orbitals belonging to the same subshell (p, d, f) does not take place until each orbital belonging to that subshell have got 1 e^- each. i.e, it is singly occupied.



4. PAULI'S EXCLUSION PRINCIPLE :

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→ No $2e^-$ in an atom can have same value of all four quantum no.s.

that is an orbital can accommodate maxm 2e-with opposite spin

eg $\rightarrow$ ① P_5

1	N	J	L
---	---	---	---

 7 X

12	12	1
----	----	---

 ✓